

# Notes on Toric Geometry

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## Abstract

These notes provide an introduction to toric geometry. The main exposition follows Chapter 7 of the book *Mirror Symmetry* [1], with supplementary material drawn from the systematic treatment in *Toric Varieties* [2]. The prerequisites for these notes include basic knowledge of algebraic geometry, for which the reader may consult Hartshorne's classic text *Algebraic Geometry* [3]. In addition, it is recommended that the reader be familiar with Chapters 1–6 of *Mirror Symmetry* [1], which develop the fundamental mathematical tools and background material necessary for the study of toric geometry and mirror symmetry.

## Contents

|          |                                                          |           |
|----------|----------------------------------------------------------|-----------|
| <b>1</b> | <b>Basic Definitions of Normal Toric Varieties</b>       | <b>2</b>  |
| 1.1      | Fans . . . . .                                           | 2         |
| 1.2      | Constructing Toric Varieties from Fans . . . . .         | 4         |
| 1.3      | Subvarieties . . . . .                                   | 6         |
| 1.4      | Compactness Criterion and Smoothness Criterion . . . . . | 7         |
| 1.5      | An Example . . . . .                                     | 7         |
| 1.6      | Constructing Fans from Toric Varieties . . . . .         | 8         |
| <b>2</b> | <b>GLSM and Intersection Numbers</b>                     | <b>9</b>  |
| 2.1      | Supersymmetric Ground States . . . . .                   | 10        |
| 2.2      | Intersection Numbers and Charges . . . . .               | 15        |
| <b>3</b> | <b>Constructions in Toric Geometry</b>                   | <b>21</b> |
| 3.1      | Orbifolds . . . . .                                      | 21        |
| 3.2      | Blow-Up . . . . .                                        | 24        |
| 3.3      | Morphisms . . . . .                                      | 30        |

|     |                                                   |    |
|-----|---------------------------------------------------|----|
| 3.4 | Local Geometry in Geometric Engineering . . . . . | 33 |
| 3.5 | Polytopes . . . . .                               | 34 |
| 3.6 | Polytopes from Toric Varieties . . . . .          | 37 |
| 3.7 | Mirror Symmetry . . . . .                         | 39 |

## 1 Basic Definitions of Normal Toric Varieties

**Definition 1.1.** A toric variety  $X$  is a complex algebraic variety containing an algebraic torus

$$T = (\mathbb{C}^*)^r$$

as a dense open set, together with an action of  $T$  on  $X$  whose restriction to  $T \subset X$  is just the usual multiplication on  $T$ .

**Definition 1.2.** A Noetherian integral domain  $R$  is called *normal* if it is integrally closed in its field of fractions, i.e. every element of  $\text{Frac}(R)$  that is integral over  $R$  already lies in  $R$ .

**Definition 1.3.** A variety  $X$  is called *normal* if it is irreducible and the local rings  $\mathcal{O}_{X,p}$  are normal for all  $p \in X$ . We will only consider the case that  $X$  is a normal variety in this section.

### 1.1 Fans

**Example 1.4.** Consider  $\mathbb{CP}^r$  with homogeneous coordinates expressed as

$$(x_1, \dots, x_{r+1}).$$

The dense open subset

$$T = \{x : x_i \neq 0, i = 1, \dots, r+1\} \subset \mathbb{CP}^r$$

is isomorphic to  $(\mathbb{C}^*)^r$  and acts on  $\mathbb{CP}^r$  by coordinatewise multiplication, giving  $\mathbb{CP}^r$  the structure of a toric variety.

**Definition 1.5.** A *lattice* is a free abelian group of finite rank, i.e. a group isomorphic to  $\mathbb{Z}^r$  for some  $r$ .

Let  $N$  be a lattice, and set

$$N_{\mathbb{R}} = N \otimes \mathbb{R}.$$

We will denote the rank of  $N$  by  $r$ . At times, we will fix an isomorphism

$$N \simeq \mathbb{Z}^r,$$

which induces an isomorphism

$$N_{\mathbb{R}} \simeq \mathbb{R}^r.$$

At other times, there will be benefits to thinking of  $N$  as an abstract lattice.

**Definition 1.6.** A *one-parameter subgroup* of a torus  $T$  is a morphism

$$\lambda: \mathbb{C}^* \rightarrow T$$

that is a group homomorphism.

For example, if  $u = (b_1, \dots, b_n) \in \mathbb{Z}^n$ , then  $u$  defines a one-parameter subgroup

$$\lambda^u: \mathbb{C}^* \rightarrow (\mathbb{C}^*)^n$$

given by

$$\lambda^u(t) = (t^{b_1}, \dots, t^{b_n}).$$

All one-parameter subgroups of  $(\mathbb{C}^*)^n$  arise in this way, and hence the group of one-parameter subgroups of  $(\mathbb{C}^*)^n$  is naturally isomorphic to  $\mathbb{Z}^n$ .

For an arbitrary torus  $T$ , its one-parameter subgroups form a free abelian group  $N$  of rank equal to the dimension of  $T$ . As with the character group, an element  $u \in N$  determines the one-parameter subgroup

$$\lambda^u: \mathbb{C}^* \rightarrow T.$$

**Definition 1.7.** A *character* of a torus  $T$  is a morphism

$$\chi: T \rightarrow \mathbb{C}^*$$

that is a group homomorphism.

For example, if  $m = (a_1, \dots, a_n) \in \mathbb{Z}^n$ , then  $m$  defines a character

$$\chi^m: (\mathbb{C}^*)^n \rightarrow \mathbb{C}^*$$

given by

$$\chi^m(t_1, \dots, t_n) = t_1^{a_1} \cdots t_n^{a_n}.$$

All characters of  $(\mathbb{C}^*)^n$  arise in this way, and hence the group of characters of  $(\mathbb{C}^*)^n$  is naturally isomorphic to  $\mathbb{Z}^n$ .

For an arbitrary torus  $T$ , its characters form a free abelian group  $M$  of rank equal to the dimension of  $T$ . It is customary to say that an element  $m \in M$  gives the character

$$\chi^m: T \rightarrow \mathbb{C}^*.$$

We will also need the accompanying vector space

$$M_{\mathbb{R}} = M \otimes \mathbb{R}.$$

**Definition 1.8** (Intrinsic pairing). Given a character  $\chi^m$  and a one-parameter subgroup  $\lambda^u$ , the composition

$$\chi^m \circ \lambda^u: \mathbb{C}^* \rightarrow \mathbb{C}^*$$

is a character of  $\mathbb{C}^*$ , hence is of the form

$$t \mapsto t^\ell$$

for some  $\ell \in \mathbb{Z}$ . We define the natural pairing between  $M$  and  $N$  by

$$\langle m, u \rangle := \ell.$$

It is easy to see  $M$  is the dual lattice

$$M = \text{Hom}(T, \mathbb{C}^*) \simeq \text{Hom}(N, \mathbb{Z}).$$

**Definition 1.9.** A *strongly convex rational polyhedral cone*

$$\sigma \subset N_{\mathbb{R}}$$

is a set of the form

$$\sigma = \{a_1 v_1 + a_2 v_2 + \cdots + a_k v_k \mid a_i \geq 0\},$$

generated by a finite set of vectors  $v_1, \dots, v_k \in N$  such that

$$\sigma \cap (-\sigma) = \{0\}.$$

Without further comment, strongly convex rational polyhedral cones will simply be referred to as *cones* in this note.

**Definition 1.10.** A *collection*  $\Sigma$  of strongly convex rational polyhedral cones in  $N_{\mathbb{R}}$  is called a *fan* if

1. each face of a cone in  $\Sigma$  is also a cone in  $\Sigma$ , and
2. the intersection of two cones in  $\Sigma$  is a face of each.

## 1.2 Constructing Toric Varieties from Fans

**Definition 1.11.** Given a cone  $\sigma \subset N_{\mathbb{R}}$ , its *dual cone* is defined by

$$\sigma^\vee = \{m \in M_{\mathbb{R}} \mid \langle m, u \rangle \geq 0 \text{ for all } u \in \sigma\}.$$

There are two standard ways to construct a toric variety  $X_\Sigma$  from a fan  $\Sigma$  yielding the same result. The original construction associates an affine toric variety

$$X_\sigma = \text{Spec } \mathbb{C}[\check{\sigma} \cap M]$$

to each cone  $\sigma \in \Sigma$ , then glues them together in a natural way to obtain  $X_\Sigma$ . We will not discuss the details of this construction here, but will recover another description of  $X_\Sigma$  later in this chapter.

Instead, it is more convenient for applications to mirror symmetry to construct toric varieties via homogeneous coordinates.

Let  $\Sigma$  be a fan in  $N_{\mathbb{R}}$ , and let  $\Sigma(1)$  denote the set of edges (one-dimensional cones) of  $\Sigma$ . For each  $\rho \in \Sigma(1)$ , let  $v_\rho \in N$  be the unique generator of the semigroup  $\rho \cap N$ . This vector  $v_\rho$  is called the *primitive generator* of  $\rho$ . Identifying  $\rho$  with  $v_\rho$ , the set  $\Sigma(1)$  may be regarded as a subset of  $N$ .

For ease of exposition, we assume that the vectors  $v_\rho$  span  $N_{\mathbb{R}}$  as a vector space for the rest of this chapter.

Let  $n = |\Sigma(1)|$ . To each edge  $\rho \in \Sigma(1)$  we associate a coordinate  $x_\rho$ . After choosing an ordering

$$\Sigma(1) = \{v_1, \dots, v_n\},$$

we may write the coordinates as

$$(x_1, \dots, x_n)$$

on  $\mathbb{C}^n$ .

**Definition 1.12.** Let  $S \subset \Sigma(1)$  be a subset which does *not* span a cone of  $\Sigma$ . Define a linear subspace  $V(S) \subset \mathbb{C}^n$  by

$$V(S) = \{x_\rho = 0 \text{ for all } \rho \in S\}.$$

Let

$$Z(\Sigma) = \bigcup_S V(S),$$

where the union is taken over all such subsets  $S \subset \Sigma(1)$ .

**Definition 1.13.** Define a homomorphism

$$\phi: \text{Hom}(\Sigma(1), \mathbb{C}^*) \longrightarrow \text{Hom}(M, \mathbb{C}^*)$$

by sending a map of sets  $f: \Sigma(1) \rightarrow \mathbb{C}^*$  to the group homomorphism

$$m \longmapsto \prod_{v \in \Sigma(1)} f(v)^{\langle m, v \rangle}.$$

In coordinates, suppose that  $v_j$  has coordinates

$$(v_{j1}, \dots, v_{jr})$$

with respect to a chosen basis of  $M$ . Then the map  $\phi$  can be written as

$$\phi: (\mathbb{C}^*)^n \longrightarrow (\mathbb{C}^*)^r, \quad (t_1, \dots, t_n) \longmapsto \left( \prod_{j=1}^n t_j^{v_{j1}}, \dots, \prod_{j=1}^n t_j^{v_{jr}} \right).$$

**Definition 1.14.** The group  $G$  is defined to be the kernel of  $\phi$ :

$$G = \text{Ker} \left( \text{Hom}(\Sigma(1), \mathbb{C}^*) \xrightarrow{\phi} \text{Hom}(M, \mathbb{C}^*) \right).$$

Since  $G \subset \text{Hom}(\Sigma(1), \mathbb{C}^*)$ , for each  $g \in G$  and each  $\rho \in \Sigma(1)$  we have  $g(v_\rho) \in \mathbb{C}^*$ . This defines an action of  $G$  on  $\mathbb{C}^n$  by

$$g \cdot (x_1, \dots, x_n) = (g(v_1)x_1, \dots, g(v_n)x_n).$$

It is easy to see that this action preserves  $\mathbb{C}^n - Z(\Sigma)$ .

**Definition 1.15.** The toric variety associated to the fan  $\Sigma$  is defined by

$$X_\Sigma = (\mathbb{C}^n - Z(\Sigma))/G.$$

The toric variety  $X_\Sigma$  contains the dense open torus

$$T = (\mathbb{C}^*)^n / G,$$

which acts on  $X_\Sigma$  by coordinatewise multiplication. It is easy to see that this torus has rank  $r$ , so that  $X_\Sigma$  is an  $r$ -dimensional toric variety.

### 1.3 Subvarieties

**Definition 1.16.** Let  $\Sigma$  be a fan and let  $\sigma \in \Sigma$  be a cone generated by edges  $\rho_1, \dots, \rho_k$ . The associated  $T$ -invariant subvariety is defined by

$$Z_\sigma = \{x \in X_\Sigma \mid x_{\rho_1} = \dots = x_{\rho_k} = 0\},$$

where  $x_\rho$  are the homogeneous coordinates of  $x$ .

**Proposition 1.17** (Classification of  $T$ -invariant subvarieties). *The assignment*

$$\sigma \mapsto Z_\sigma$$

*defines an order-reversing bijection between cones in the fan  $\Sigma$  and non-empty  $T$ -invariant subvarieties of  $X_\Sigma$ .*

**Remark 1.18.** In particular, the edges of  $\Sigma$  are in one-to-one correspondence with  $T$ -invariant divisors in  $X_\Sigma$ . More generally, if  $\sigma$  is a  $k$ -dimensional cone, then  $Z_\sigma$  is an  $(r - k)$ -dimensional subvariety of  $X_\Sigma$ .

**Remark 1.19.** If a set of edges  $\{\rho_1, \dots, \rho_\ell\}$  does not span a cone in  $\Sigma$ , then the equations

$$x_{\rho_1} = \dots = x_{\rho_\ell} = 0$$

define a subset of  $\mathbb{C}^n$  contained in  $Z(\Sigma)$ , hence they define the empty set in  $X_\Sigma$ .

**Remark 1.20.** Each  $Z_\sigma$  is itself a toric variety. Its fan is constructed by replacing  $N$  with the quotient lattice

$$N' = N / \langle \sigma \cap N \rangle,$$

and projecting each cone in  $\Sigma$  which contains  $\sigma$  as a face to  $N'$ . The resulting collection of cones forms a fan in  $N'$ .

At this point, we relate the homogeneous coordinate construction with the standard affine covering

$$X_\Sigma = \bigcup_{\sigma \in \Sigma} X_\sigma.$$

**Definition 1.21.** Let  $\sigma \in \Sigma$  be an  $r$ -dimensional cone. The corresponding affine open subset  $X_\sigma \subset X_\Sigma$  is defined as the subset obtained by setting

$$x_\rho = 1$$

for all  $\rho \in \Sigma(1)$  which are not edges of  $\sigma$ .

**Remark 1.22.** Let  $\Sigma_\sigma \subset \Sigma$  be the subfan consisting of  $\sigma$  and all of its faces. Then one checks directly that

$$X_{\Sigma_\sigma} \simeq X_\sigma.$$

This description is particularly useful for local calculations on  $X_\Sigma$ .

## 1.4 Compactness Criterion and Smoothness Criterion

**Proposition 1.23** (Compactness criterion for toric varieties). *Let  $X_\Sigma$  be a toric variety. Then the following are equivalent:*

1.  $X_\Sigma$  is compact in the classical topology.
2. The fan  $\Sigma$  is complete, i.e.,

$$|\Sigma| = \bigcup_{\sigma \in \Sigma} \sigma = N_{\mathbb{R}}.$$

**Proposition 1.24** (Smoothness criterion for toric varieties). *A toric variety  $X_\Sigma$  is smooth if and only if every cone  $\sigma \in \Sigma$  is generated by a  $\mathbb{Z}$ -basis of the lattice*

$$\text{Span}_{\mathbb{R}}(\sigma) \cap N.$$

*Proof.* Suppose that each cone  $\sigma \in \Sigma$  is generated by a  $\mathbb{Z}$ -basis of  $\text{Span}_{\mathbb{R}}(\sigma) \cap N$ . Let  $\sigma$  be a top-dimensional cone. Then the associated affine toric variety  $X_{\Sigma_\sigma}$  has trivial group  $G$ , and hence

$$X_{\Sigma_\sigma} \simeq X_\sigma \simeq \mathbb{C}^r.$$

Therefore  $X_\Sigma$  is locally smooth, and hence smooth.

Conversely, assume that  $X_\Sigma$  is smooth. For a cone  $\sigma \in \Sigma$ , the affine toric variety

$$X_\sigma = \text{Spec } \mathbb{C}[\tilde{\sigma} \cap M]$$

is smooth at the origin if and only if the maximal ideal at the origin is generated by  $r$  elements. These generators correspond to generators of the dual cone  $\tilde{\sigma}$ , which implies that  $\sigma$  is generated by a  $\mathbb{Z}$ -basis of  $\text{Span}_{\mathbb{R}}(\sigma) \cap N$ . The details are left to the reader.  $\square$

## 1.5 An Example

**Example 1.25.** We construct the total space  $\mathcal{O}_{\mathbb{CP}^2}(-3)$  from a fan  $\Sigma$  in  $\mathbb{R}^3$ . Set

$$\Sigma(1) = \{(1, 0, 1), (0, 1, 1), (-1, -1, 1), (0, 0, 1)\}.$$

The convex hull of  $\Sigma(1)$  is a triangle in the plane  $z = 1$  with vertices  $\{v_1, v_2, v_3\}$ ; the vector  $v_4$  lies in the interior of this triangle, subdividing it into three smaller triangles. The three-dimensional cones in  $\Sigma$  are the cones over these triangles, and the remaining cones are their faces. These cones do not span  $\mathbb{R}^3$ , which is consistent with the fact that  $\mathcal{O}_{\mathbb{CP}^2}(-3)$  is non-compact (non-compact toric varieties of this type are useful in geometric engineering).

We compute that

$$Z(\Sigma) = \{x_1 = x_2 = x_3 = 0\}.$$

Moreover, the group  $G$  is the kernel of the homomorphism

$$\phi: (\mathbb{C}^*)^4 \longrightarrow (\mathbb{C}^*)^3, \quad (t_1, t_2, t_3, t_4) \longmapsto (t_1 t_3^{-1}, t_2 t_3^{-1}, t_1 t_2 t_3 t_4),$$

so that

$$G = \{(t, t, t, t^{-3}) \mid t \in \mathbb{C}^*\} \simeq \mathbb{C}^*.$$

There are four  $T$ -invariant divisors  $D_1, \dots, D_4$ . Since  $\{v_1, v_2, v_3\}$  does not span a cone in  $\Sigma$ , the divisors  $D_1, D_2, D_3$  have empty intersection. All other triples of divisors intersect in a point, since the corresponding edges span a three-dimensional cone. The toric description of the  $D_i$  shows that  $D_4$  is compact, while the other divisors are non-compact.

Projection to the first three factors defines a morphism

$$X_\Sigma \longrightarrow \mathbb{CP}^2$$

whose fibers are isomorphic to  $\mathbb{C}$ . Hence  $X_\Sigma$  is a line bundle over  $\mathbb{CP}^2$ , as claimed. The divisor  $D_4$  is identified with the zero section of the bundle, and the divisors  $D_1, D_2, D_3$  are the restrictions of this bundle over the coordinate lines  $x_1 = 0$ ,  $x_2 = 0$ ,  $x_3 = 0$ .

## 1.6 Constructing Fans from Toric Varieties

Let  $X$  be a normal toric variety containing the torus

$$T \simeq (\mathbb{C}^*)^r.$$

We consider the lattice

$$N = \text{Hom}(\mathbb{C}^*, T) \simeq \mathbb{Z}^r,$$

whose elements are one-parameter subgroups of  $T$ . Identifying  $T$  with  $(\mathbb{C}^*)^r$ , this identification may be written as

$$\mathbb{Z}^r \simeq N, \quad (a_1, \dots, a_r) \longmapsto (t \mapsto (t^{a_1}, \dots, t^{a_r})).$$

**Definition 1.26.** Let  $\psi \in N$  be a one-parameter subgroup. Define a morphism

$$f: \mathbb{C}^* \longrightarrow X, \quad f(t) = \psi(t) \cdot 1,$$

where 1 denotes the identity element of  $T \subset X$ . If the limit

$$\lim_{t \rightarrow 0} f(t)$$

exists in  $X$ , we define the associated  $T$ -invariant subvariety

$$Z_\psi = \overline{T \cdot \lim_{t \rightarrow 0} f(t)}.$$

**Definition 1.27.** Let  $N' \subset N$  be the set of one-parameter subgroups  $\psi$  for which  $Z_\psi$  exists. On  $N'$  we define an equivalence relation by

$$\psi \sim \psi' \iff Z_\psi = Z_{\psi'}.$$

**Definition 1.28.** For a fixed equivalence class  $[\psi]$ , define the associated cone  $\sigma$  to be the closure of the convex hull in  $N_\mathbb{R}$  of all one-parameter subgroups in  $[\psi]$ .

**Definition 1.29.** The collection of all cones obtained in this way forms a fan  $\Sigma$  in  $N_\mathbb{R}$ .

**Proposition 1.30.** *Let  $X$  be a normal toric variety with torus  $T$ . The fan  $\Sigma$  constructed above satisfies*

$$X \simeq X_\Sigma.$$

**Remark 1.31.** Let  $\psi \in N$  be contained in a cone of  $\Sigma$ . Then there exists a unique cone  $\sigma \in \Sigma$  such that  $\psi$  lies in the relative interior of  $\sigma$ . For this cone,

$$Z_\psi = Z_\sigma.$$

**Remark 1.32** (Compactness criterion revisited). This construction gives a conceptual explanation of the compactness criterion for toric varieties. If

$$|\Sigma| \subsetneq N_{\mathbb{R}},$$

then there exists a one-parameter subgroup  $\psi \in N$  not contained in any cone of  $\Sigma$ . For such a  $\psi$ , the limit

$$\lim_{t \rightarrow 0} \psi(t)$$

does not exist in  $X_\Sigma$ , and hence  $X_\Sigma$  cannot be compact.

**Example 1.33** (Recovering the fan of  $\mathbb{CP}^2$  from one-parameter subgroups). We revisit  $\mathbb{CP}^2$  as a toric variety and recover its fan using the construction of one-parameter subgroups.

Using the rescaling of homogeneous coordinates, we may normalize the first coordinate to 1, identifying the dense torus as

$$T = \{(1, t_1, t_2) \mid t_i \in \mathbb{C}^*\} \simeq (\mathbb{C}^*)^2.$$

With this identification, the torus action on  $\mathbb{CP}^2$  is given by

$$(t_1, t_2) \cdot (x_1, x_2, x_3) = (x_1, t_1 x_2, t_2 x_3).$$

The one-parameter subgroups of  $T$  are indexed by  $(a, b) \in \mathbb{Z}^2$  and are given by

$$\psi_{a,b}(t) = (t^a, t^b) \in N = \text{Hom}(\mathbb{C}^*, T).$$

Using the embedding  $T \subset \mathbb{CP}^2$ , we may study the limit

$$\lim_{t \rightarrow 0} \psi_{a,b}(t) \in \mathbb{CP}^2.$$

There are seven possibilities for the limit points and their orbit closures, summarized in Table 1. The closures of the regions defined by the conditions on  $(a, b)$  determine the cones of the fan of  $\mathbb{CP}^2$ . In this way, we recover the fan consisting of seven cones, in agreement with the standard toric description of  $\mathbb{CP}^2$ . This also recovers the correspondence between cones and non-empty  $T$ -invariant subvarieties.

## 2 GLSM and Intersection Numbers

The *gauged linear sigma model* (GLSM) is a two-dimensional gauge theory. For present purposes, we restrict our attention to GLSMs without a superpotential.

| Condition on $(a, b)$ | $\lim_{t \rightarrow 0} \psi(t)$ | orbit closure   |
|-----------------------|----------------------------------|-----------------|
| $a > 0, b > 0$        | $(1, 0, 0)$                      | $\{(1, 0, 0)\}$ |
| $a < 0, b > a$        | $(0, 1, 0)$                      | $\{(0, 1, 0)\}$ |
| $b < 0, b < a$        | $(0, 0, 1)$                      | $\{(0, 0, 1)\}$ |
| $a = b < 0$           | $(0, 1, 1)$                      | $\{x_1 = 0\}$   |
| $a > 0, b = 0$        | $(1, 0, 1)$                      | $\{x_2 = 0\}$   |
| $a = 0, b > 0$        | $(1, 1, 0)$                      | $\{x_3 = 0\}$   |
| $a = b = 0$           | $(1, 1, 1)$                      | $\mathbb{CP}^2$ |

Table 1: Limits of one-parameter subgroups in  $\mathbb{CP}^2$

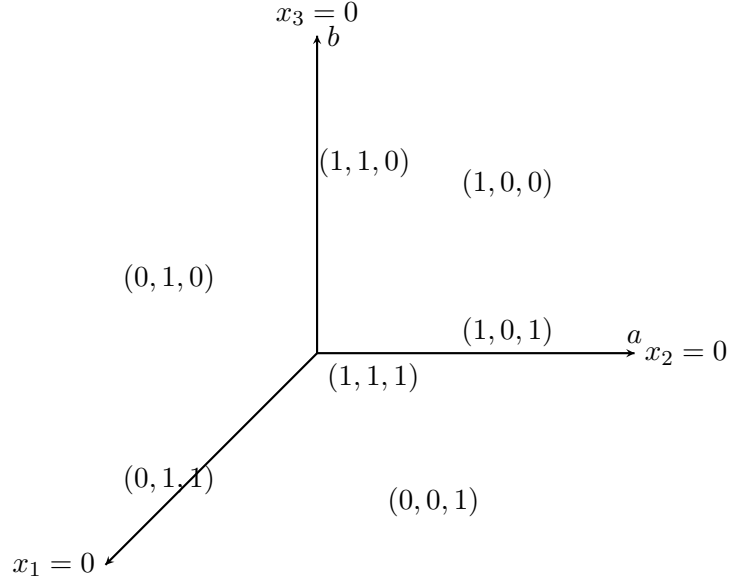


Figure 1: One-parameter subgroups and limit points for  $\mathbb{CP}^2$

## 2.1 Supersymmetric Ground States

**Definition 2.1** (GLSM without superpotential). We consider a two-dimensional  $\mathcal{N} = (2, 2)$  gauged linear sigma model (GLSM) with gauge group  $U(1)^s$ , vector superfields  $V_1, \dots, V_s$ , and  $n$  chiral superfields  $\Phi_1, \dots, \Phi_n$ .

The charge of  $\Phi_i$  under the  $a$ -th  $U(1)$  factor is denoted by  $Q_{i,a} \in \mathbb{Z}$ , and the scalar component of  $\Phi_i$  is denoted by  $\phi_i$ . The Lagrangian takes the form

$$L = L_{\text{kin}} + L_{\text{gauge}} + L_{D,\theta},$$

where the three terms correspond respectively to the kinetic energy of the chiral fields, the kinetic energy of the gauge fields, and the Fayet–Iliopoulos (FI) and theta terms.

The scalar potential is given by

$$U(\phi) = \sum_{a=1}^s \frac{e_a^2}{2} \left( \sum_{i=1}^n Q_{i,a} |\phi_i|^2 - r_a \right)^2,$$

where  $e_a$  are the gauge couplings and  $r_a \in \mathbb{R}$  are the FI parameters.

The supersymmetric ground states are defined as the solutions of the D-term equations

$$\sum_{i=1}^n Q_{i,a} |\phi_i|^2 = r_a, \quad a = 1, \dots, s,$$

modulo the action of the gauge group  $U(1)^s$ .

**Theorem 2.2** (Vacuum moduli space as a toric variety). *For a GLSM with gauge group  $U(1)^s$  and charge matrix  $Q_{i,a}$ , and for an appropriate choice of FI parameters  $r_a$ , the space of supersymmetric ground states is an  $(n-s)$ -dimensional normal toric variety  $X_\Sigma$  whose fan  $\Sigma$  has  $n$  one-dimensional cones.*

*Proof.* Define a subgroup  $G = (\mathbb{C}^*)^s \subset (\mathbb{C}^*)^n$  by the embedding

$$(t_1, \dots, t_s) \mapsto \left( \prod_{a=1}^s t_a^{Q_{1,a}}, \dots, \prod_{a=1}^s t_a^{Q_{n,a}} \right).$$

The residual torus acting effectively on the vacuum moduli space is

$$T = (\mathbb{C}^*)^n / G.$$

Let

$$N = \text{Hom}(\mathbb{C}^*, T)$$

be the lattice of one-parameter subgroups of the torus  $T$ . There exists a collection of elements

$$S = \{v_1, \dots, v_n\} \subset N$$

such that replacing the set  $\Sigma(1)$  of rays in the Cox construction by  $S$  reproduces the embedding. For simplicity, we assume the  $v_i$  are distinct and set

$$\Sigma(1) := S.$$

Consider subsets

$$P = \{v_{i_1}, \dots, v_{i_k}\} \subset \Sigma(1)$$

such that there are no solutions to the D-term equations

$$\sum_i Q_{i,a} |\phi_i|^2 = r_a$$

with  $\phi_{i_1} = \dots = \phi_{i_k} = 0$ .

Then there exists a unique fan  $\Sigma$  whose one-dimensional cones are  $\Sigma(1)$  and such that a subset of  $\Sigma(1)$  fails to span a cone of  $\Sigma$  if and only if it is one of the forbidden subsets  $P$ , which is just the fan of the space of supersymmetric ground states.  $\square$

**Remark 2.3.** We explain how the gauge charges in a GLSM determine the toric variety of supersymmetric ground states, and conversely how the toric data recover the GLSM charges.

One associates to the GLSM a set of vectors

$$\Sigma(1) = \{v_1, \dots, v_n\}$$

which generate the one-dimensional cones of a fan  $\Sigma$ . By construction, the charges encode the linear relations among the  $v_i$ :

$$\sum_{i=1}^n Q_{i,a} v_i = 0, \quad a = 1, \dots, s.$$

Conversely, starting from the set of toric rays  $\Sigma(1)$ , one considers the lattice

$$\Lambda := \left\{ (q_1, \dots, q_n) \in \mathbb{Z}^n \left| \sum_{i=1}^n q_i v_i = 0 \right. \right\},$$

which is the lattice of all  $\mathbb{Z}$ -linear relations among the  $v_i$ . This lattice has rank  $s$ .

Choosing a basis of  $\Lambda$ , one obtains relations

$$\sum_{i=1}^n Q'_{i,a} v_i = 0, \quad a = 1, \dots, s,$$

and hence a charge matrix  $Q'_{i,a}$ .

From linear algebra, the coefficients  $Q'_{i,a}$  are precisely the charges of the original chiral superfields  $\Phi_i$ , with the understanding that different choices of basis of  $\Lambda$  correspond to different ways of writing the same gauge group as a product of  $s$  copies of  $U(1)$ .

**Remark 2.4** (Kempf–Ness correspondence for GLSM). In the GLSM construction, the space of supersymmetric ground states is defined as the Kähler quotient

$$\mathcal{M}_{\text{vac}} = \mu^{-1}(r)/U(1)^s,$$

where  $\mu : \mathbb{C}^n \rightarrow \mathbb{R}^s$  is the moment map

$$\mu_a(\phi) = \sum_{i=1}^n Q_{i,a} |\phi_i|^2,$$

and  $r = (r_1, \dots, r_s)$  are the FI parameters. Note that the complexified gauge group  $G = (\mathbb{C}^*)^s$  does *not* preserve the level set  $\mu^{-1}(r)$ .

Nevertheless, one can equivalently describe the vacuum moduli space as a GIT quotient

$$\mathcal{M}_{\text{vac}} \cong (\mathbb{C}^n \setminus Z_r)/G,$$

where  $Z_r$  is the unstable locus determined by the FI parameters. This equivalence is a special case of the Kempf–Ness correspondence, which asserts that every  $G$ -orbit in  $\mathbb{C}^n \setminus Z_r$  intersects  $\mu^{-1}(r)$  in a unique  $U(1)^s$ -orbit. In particular, the  $G$ -orbit closure contains a unique representative minimizing the norm, and this representative satisfies the D-term equations.

Thus, although the  $G$ -action does not preserve  $\mu^{-1}(r)$ , the Kähler quotient by  $U(1)^s$  is canonically identified with the GIT quotient by  $G$ . This identification allows one to construct the vacuum moduli space as a toric variety using the holomorphic quotient  $(\mathbb{C}^n \setminus Z_r)/G$ .

**Remark 2.5.** The space of supersymmetric ground states depends crucially on the choice of FI parameters  $r_a$ . For certain values of  $r_a$ , the vacuum moduli space may fail to be a toric variety, or may even be empty.

For example, consider a  $U(1)$  gauge theory with two chiral fields of charges  $(1, 1)$ . The D-term equation

$$|\phi_1|^2 + |\phi_2|^2 = r$$

has no solutions if  $r < 0$ , so in this regime the space of supersymmetric ground states is empty. This shows that the existence of a geometric phase already imposes nontrivial constraints on the FI parameters.

More generally, the dependence of the vacuum geometry on the FI parameters is encoded in the *GKZ decomposition* of the parameter space, which describes a chamber structure corresponding to different geometric (and non-geometric) phases of the theory. *GKZ decomposition* describes how the FI parameter space is divided into chambers, each of which corresponds to a distinct toric (or non-geometric) phase of the GLSM. Crossing a wall in the GKZ fan corresponds to a birational transformation of the target space, such as a blow-up or flop.

Finally, we note that for a general toric variety the group  $G$  appearing in the GLSM construction need not be a pure torus  $(\mathbb{C}^*)^s$ ; it may also contain finite abelian factors. In such cases, the resulting vacuum moduli space is a toric *orbifold*, rather than a smooth toric variety. Producing these geometries requires allowing orbifold phases in the GLSM, which will be discussed later.

**Example 2.6** (A  $U(1)$  GLSM with charges  $(1, 1, 1, -3)$ ). Consider a two-dimensional  $U(1)$  gauge theory with four chiral fields  $\Phi_1, \Phi_2, \Phi_3, \Phi_4$  of charges

$$Q = (1, 1, 1, -3).$$

Let  $r \in \mathbb{R}$  be the FI parameter. The D-term (moment map) equation is

$$|\phi_1|^2 + |\phi_2|^2 + |\phi_3|^2 - 3|\phi_4|^2 = r,$$

where  $\phi_i$  denotes the scalar component of  $\Phi_i$ .

**(i) The phase  $r > 0$ .** If  $r > 0$ , then D-term equation implies that we cannot have  $\phi_1 = \phi_2 = \phi_3 = 0$ . Equivalently, in the holomorphic/GIT picture, the unstable locus contains  $\{\phi_1 = \phi_2 = \phi_3 = 0\}$ , hence the quotient removes this locus. In this chamber one obtains the geometric phase whose vacuum moduli space is the total space of the line bundle

$$\text{Tot}(\mathcal{O}_{\mathbb{P}^2}(-3)).$$

(Concretely, this is the toric variety described by the fan in the earlier toric example, where the excluded set is  $Z = \{x_1 = x_2 = x_3 = 0\}$ .)

**(ii) The phase  $r < 0$  and the orbifold  $\mathbb{C}^3/\mathbb{Z}_3$ .** If  $r < 0$ , then D-term equation forces  $\phi_4 \neq 0$  (otherwise the left-hand side is  $\geq 0$ ). This has two closely related consequences: (a) *Gauge-fixing and the residual  $\mathbb{Z}_3$* . The  $U(1)$  action is

$$(\phi_1, \phi_2, \phi_3, \phi_4) \mapsto (e^{i\theta}\phi_1, e^{i\theta}\phi_2, e^{i\theta}\phi_3, e^{-3i\theta}\phi_4).$$

Since  $\phi_4 \neq 0$ , we may use the continuous  $U(1)$ -freedom to fix the phase of  $\phi_4$  (and in the holomorphic picture, to set  $\phi_4 = 1$ ). However, the gauge transformation with  $\theta = 2\pi/3$  preserves this gauge

choice because  $e^{-3i\theta} = e^{-2\pi i} = 1$ . Therefore, after gauge-fixing there remains a *finite residual gauge group*

$$\mathbb{Z}_3 = \{e^{2\pi i k/3} \mid k = 0, 1, 2\} \subset U(1).$$

This residual  $\mathbb{Z}_3$  acts on  $(\phi_1, \phi_2, \phi_3)$  by simultaneous multiplication by a primitive cube root of unity:

$$(\phi_1, \phi_2, \phi_3) \mapsto (\omega\phi_1, \omega\phi_2, \omega\phi_3), \quad \omega = e^{2\pi i/3}.$$

Hence the moduli space in this phase is

$$\mathbb{C}^3/\mathbb{Z}_3.$$

(b) *The same statement in the holomorphic/GIT quotient language.* Consider the holomorphic  $(\mathbb{C}^*)$ -action with weights  $(1, 1, 1, -3)$ :

$$\lambda \cdot (\phi_1, \phi_2, \phi_3, \phi_4) = (\lambda\phi_1, \lambda\phi_2, \lambda\phi_3, \lambda^{-3}\phi_4).$$

For  $r < 0$ , stability forces  $\phi_4 \neq 0$ , so on the stable locus we set  $\phi_4 = 1$  by choosing  $\lambda = \phi_4^{1/3}$ . The ambiguity of the cube root is exactly a  $\mathbb{Z}_3$ -freedom, producing the same quotient  $\mathbb{C}^3/\mathbb{Z}_3$ .

(iii) **Relation between the two phases.** Crossing  $r = 0$  changes the GIT stability condition (a wall-crossing in the GKZ/secondary fan). Geometrically, the  $r > 0$  phase gives the smooth resolution  $\text{Tot}(\mathcal{O}_{\mathbb{P}^2}(-3))$ , while the  $r < 0$  phase is the orbifold  $\mathbb{C}^3/\mathbb{Z}_3$ ; the former can be viewed as a (crepant) resolution of the latter.

**Definition 2.7** (The  $P \mid Q$  matrix). Let  $\Sigma(1) = \{v_1, \dots, v_n\}$  be the set of generators of the one-dimensional cones of a toric fan, and let

$$\Lambda := \left\{ (q_1, \dots, q_n) \in \mathbb{Z}^n \mid \sum_{i=1}^n q_i v_i = 0 \right\}$$

be the lattice of  $\mathbb{Z}$ -linear relations among the  $v_i$ , of rank  $s$ .

Choosing a basis of the lattice  $N$  and a basis of  $\Lambda$ , the toric data can be encoded in a matrix

$$P \mid Q,$$

where the matrix  $P$  has row vectors given by the generators  $v_i$  of the toric rays; the matrix  $Q$  has column vectors forming a basis of the relation lattice  $\Lambda$ .

Equivalently, the matrix  $Q$  satisfies the relations

$$\sum_{i=1}^n Q_{i,a} v_i = 0, \quad a = 1, \dots, s.$$

**Example 2.8** (The Hirzebruch surface  $F_n$  as a GLSM vacuum). Consider the Hirzebruch surface  $F_n$ . The lattice of  $\mathbb{Z}$ -linear relations among the toric rays is generated by the matrix

$$P \mid Q = \left( \begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ -1 & -n & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & -n \end{array} \right).$$

It follows that this toric variety can be realized as the space of supersymmetric ground states of a GLSM with gauge group  $U(1)^2$  and four chiral superfields. The corresponding charge vectors are

$$(0, 1), \quad (0, 1), \quad (1, 0), \quad (1, -n),$$

which can be checked directly from the relation lattice.

## 2.2 Intersection Numbers and Charges

**Definition 2.9** (*T*-invariant divisors). Let  $X_\Sigma$  be a toric variety with fan  $\Sigma$ . For each  $\rho \in \Sigma(1)$ , denote by  $D_\rho$  the *T*-invariant divisor corresponding to the ray  $\rho$ . Then the group of *T*-invariant Weil divisors is

$$\mathbb{Z}^{\Sigma(1)} \cong \bigoplus_{\rho \in \Sigma(1)} \mathbb{Z} \cdot D_\rho.$$

**Definition 2.10** (Principal divisors from characters). Let  $M$  be the character lattice of the torus  $T$ . Each character  $m \in M$  defines a rational function on  $X_\Sigma$ , whose zeros and poles determine a principal divisor

$$(m) = \sum_{\rho \in \Sigma(1)} \langle m, v_\rho \rangle D_\rho.$$

This defines a homomorphism

$$M \longrightarrow \mathbb{Z}^{\Sigma(1)}.$$

This definition above is just equivalent to the usual divisor we use in algebraic geometry. We will quickly prove this.

**Lemma 2.11** (Generic point lies in a toric affine chart). *Let  $X_\Sigma$  be a toric variety,  $\rho \in \Sigma(1)$ , and  $\sigma \in \Sigma$  such that  $\rho \subset \sigma$ . Let  $D_\rho \subset X_\Sigma$  be the torus-invariant prime divisor corresponding to  $\rho$ , and let  $\eta_\rho$  be its generic point. Then  $\eta_\rho \in U_\sigma$  and*

$$\mathcal{O}_{X_\Sigma, \eta_\rho} \cong \mathcal{O}_{U_\sigma, \eta_\rho}.$$

*Proof.* Since  $\rho \subset \sigma$ , the orbit–face correspondence implies that  $D_\rho \cap U_\sigma$  is the orbit closure  $V(\rho) \subset U_\sigma$ , hence it is a nonempty irreducible closed subset of  $U_\sigma$ . The generic point  $\eta_\rho$  of  $D_\rho$  belongs to every nonempty open subset of  $D_\rho$ ; in particular, it lies in the nonempty open subset  $D_\rho \cap U_\sigma \subset D_\rho$ . Therefore  $\eta_\rho \in U_\sigma$ . The identification of local rings follows from the general fact that for an open immersion  $j : U_\sigma \hookrightarrow X_\Sigma$  and a point  $x \in U_\sigma$ , one has  $\mathcal{O}_{X_\Sigma, x} \cong \mathcal{O}_{U_\sigma, x}$ .  $\square$

**Lemma 2.12** (The divisor  $D_\rho$  on an affine chart). *Let  $\sigma \in \Sigma$  be a cone and  $\rho \preceq \sigma$  a ray. Write  $U_\sigma = \text{Spec } \mathbb{C}[\sigma^\vee \cap M]$ . Then the closed subscheme  $D_\rho \cap U_\sigma \subset U_\sigma$  is defined by the prime ideal*

$$\mathfrak{p}_\rho(\sigma) := \left( \chi^u \mid u \in \sigma^\vee \cap M, \langle u, v_\rho \rangle > 0 \right) \subset \mathbb{C}[\sigma^\vee \cap M].$$

*In particular,  $D_\rho \cap U_\sigma$  is nonempty and irreducible.*

*Proof.* This is the standard orbit–face description of toric subvarieties: for a face  $\tau \preceq \sigma$ , the orbit closure  $V(\tau) \subset U_\sigma$  is defined by the ideal generated by monomials  $\chi^u$  such that  $\langle u, v \rangle > 0$  for some (equivalently any)  $v \in \tau$  in the relative interior. For  $\tau = \rho$  this yields the stated ideal.  $\square$

**Proposition 2.13** (Local coordinate for  $D_\rho$  in the smooth case). *Assume  $X_\Sigma$  is smooth. Let  $\sigma \in \Sigma$  be a maximal cone of dimension  $r$  with rays  $\rho_1, \dots, \rho_r$  and primitive generators  $v_{\rho_1}, \dots, v_{\rho_r} \in N$ . Then  $\{v_{\rho_i}\}$  is a  $\mathbb{Z}$ -basis of  $N$ , and letting  $\{u_i\} \subset M$  be the dual basis  $\langle u_i, v_{\rho_j} \rangle = \delta_{ij}$ , we have an identification*

$$\mathbb{C}[\sigma^\vee \cap M] \cong \mathbb{C}[\chi^{u_1}, \dots, \chi^{u_r}] \cong \mathbb{C}[x_1, \dots, x_r], \quad x_i := \chi^{u_i}.$$

Under this identification, for each  $j$ ,

$$D_{\rho_j} \cap U_\sigma = V(x_j), \quad \text{and} \quad \mathcal{O}_{U_\sigma, \eta_{\rho_j}} \cong \mathbb{C}[x_1, \dots, x_r]_{(x_j)},$$

so  $x_j$  is a uniformizer of the DVR  $\mathcal{O}_{X_\Sigma, \eta_{\rho_j}}$ .

*Proof.* Smoothness implies  $\sigma$  is unimodular, hence the primitive ray generators of a maximal cone form a  $\mathbb{Z}$ -basis of  $N$  and admit a dual basis in  $M$ . Then  $\sigma^\vee \cap M$  is the free commutative monoid generated by  $u_1, \dots, u_r$ , so the semigroup algebra is a polynomial algebra. The description of  $D_{\rho_j} \cap U_\sigma$  as  $V(x_j)$  follows by comparing the ideal  $\mathfrak{p}_{\rho_j}(\sigma)$  from the previous lemma with the condition  $\langle u_i, v_{\rho_j} \rangle = \delta_{ij}$ : a monomial  $\chi^u = \prod_i x_i^{a_i}$  lies in  $\mathfrak{p}_{\rho_j}(\sigma)$  iff  $\langle u, v_{\rho_j} \rangle = a_j > 0$ , i.e. iff it is divisible by  $x_j$ , so  $\mathfrak{p}_{\rho_j}(\sigma) = (x_j)$ . Localizing at the generic point gives the stated DVR.  $\square$

**Theorem 2.14** (Valuation of a character along  $D_\rho$ ). *Let  $X_\Sigma$  be a smooth toric variety. For any  $m \in M$  and any ray  $\rho \in \Sigma(1)$ ,*

$$\text{ord}_{D_\rho}(\chi^m) = \langle m, v_\rho \rangle.$$

*Consequently, the principal divisor of  $\chi^m$  is*

$$\div(\chi^m) = \sum_{\rho \in \Sigma(1)} \langle m, v_\rho \rangle D_\rho.$$

*Proof.* Fix a maximal cone  $\sigma$  containing  $\rho$  (such a  $\sigma$  exists) and write  $\rho = \rho_j$  in the notation of the previous proposition. By Lemma 1,  $\eta_\rho \in U_\sigma$  and  $\mathcal{O}_{X_\Sigma, \eta_\rho} \cong \mathcal{O}_{U_\sigma, \eta_\rho}$ . By the proposition, we may identify

$$\mathcal{O}_{U_\sigma, \eta_\rho} \cong \mathbb{C}[x_1, \dots, x_r]_{(x_j)}$$

with uniformizer  $x_j = \chi^{u_j}$ . Writing  $m = \sum_{i=1}^r a_i u_i$  in the dual basis, we have

$$\chi^m = \prod_{i=1}^r (\chi^{u_i})^{a_i} = \prod_{i=1}^r x_i^{a_i}.$$

In the DVR  $\mathbb{C}[x_1, \dots, x_r]_{(x_j)}$ , all  $x_i$  with  $i \neq j$  are units, hence the valuation  $\text{ord}_{D_\rho}(\chi^m)$  equals the exponent of the uniformizer  $x_j$ , namely  $a_j$ . By duality,  $a_j = \langle m, v_{\rho_j} \rangle = \langle m, v_\rho \rangle$ . This proves the first formula. The expression for  $\div(\chi^m)$  follows from the definition

$$\div(f) = \sum_{\rho \in \Sigma(1)} \text{ord}_{D_\rho}(f) D_\rho$$

applied to  $f = \chi^m$ .  $\square$

**Remark 2.15.** For  $v \in N$  let  $\lambda_v : \mathbb{C}^* \rightarrow T$  be the associated one-parameter subgroup, and for  $m \in M$  let  $\chi^m : T \rightarrow \mathbb{C}^*$  be the character. Then  $\chi^m \circ \lambda_v(t) = t^{\langle m, v \rangle}$ . In particular, taking  $v = v_\rho$  recovers the same integer that appears in  $\text{ord}_{D_\rho}(\chi^m) = \langle m, v_\rho \rangle$ .

**Theorem 2.16** (Linear equivalence equals homological equivalence for smooth projective toric varieties). *Let  $X_\Sigma$  be a smooth projective toric variety, and let  $D, D'$  be divisors on  $X_\Sigma$ . Then the following are equivalent:*

1.  $D$  and  $D'$  are linearly equivalent;

2.  $\mathcal{O}(D) \cong \mathcal{O}(D')$ ;
3.  $[D] = [D'] \in H_{2n-2}(X_\Sigma, \mathbb{Z})$ ;
4.  $D - D' = (m)$  for some  $m \in M$ .

*Proof.* We prove the only nontrivial implication  $(3) \Rightarrow (1)$ , since the other implications

$$(4) \Leftrightarrow (1) \Rightarrow (2) \Rightarrow (3)$$

hold for divisors on any normal variety.

Assume that  $[D] = [D'] \in H_{2n-2}(X_\Sigma, \mathbb{Z})$ . By Poincaré duality, this is equivalent to

$$\text{PD}([D]) = \text{PD}([D']) \in H^2(X_\Sigma, \mathbb{Z}).$$

For any divisor  $E$  on a smooth variety, the Poincaré dual of its homology class is the first Chern class of the associated line bundle:

$$\text{PD}([E]) = c_1(\mathcal{O}(E)).$$

Hence

$$c_1(\mathcal{O}(D)) = c_1(\mathcal{O}(D')) \in H^2(X_\Sigma, \mathbb{Z}).$$

We now use the following standard fact for smooth projective toric varieties.

**Lemma 2.17** (Vanishing and Picard–cohomology isomorphism for toric varieties). *Let  $X_\Sigma$  be a smooth projective toric variety. Then*

$$H^i(X_\Sigma, \mathcal{O}_{X_\Sigma}) = 0 \quad \text{for all } i > 0,$$

*and the first Chern class map induces an isomorphism*

$$c_1 : \text{Pic}(X_\Sigma) \xrightarrow{\sim} H^2(X_\Sigma, \mathbb{Z}).$$

*Proof.* For smooth projective toric varieties, there is a highly nontrivial fact that the Hodge numbers satisfy

$$H^{p,q}(X_\Sigma) = 0 \quad \text{for } p \neq q.$$

In particular,

$$H^i(X_\Sigma, \mathcal{O}_{X_\Sigma}) = H^{0,i}(X_\Sigma) = 0 \quad (i > 0).$$

Applying cohomology to the exponential exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_{X_\Sigma} \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}_{X_\Sigma}^* \longrightarrow 0$$

gives the exact segment

$$H^1(\mathcal{O}_{X_\Sigma}) \longrightarrow \text{Pic}(X_\Sigma) \xrightarrow{c_1} H^2(X_\Sigma, \mathbb{Z}) \longrightarrow H^2(\mathcal{O}_{X_\Sigma}).$$

The vanishing of  $H^1(\mathcal{O}_{X_\Sigma})$  and  $H^2(\mathcal{O}_{X_\Sigma})$  implies that  $c_1$  is an isomorphism. □

Returning to the theorem, since  $c_1 : \text{Pic}(X_\Sigma) \rightarrow H^2(X_\Sigma, \mathbb{Z})$  is an isomorphism, the equality

$$c_1(\mathcal{O}(D)) = c_1(\mathcal{O}(D'))$$

implies

$$\mathcal{O}(D) \cong \mathcal{O}(D').$$

Therefore  $D$  and  $D'$  are linearly equivalent, completing the proof.  $\square$

**Proposition 2.18** (Divisor class exact sequence). *Every divisor on  $X_\Sigma$  is linearly equivalent to a  $T$ -invariant divisor. Consequently, there is an exact sequence*

$$0 \longrightarrow M \longrightarrow \mathbb{Z}^{\Sigma(1)} \longrightarrow A_{r-1}(X_\Sigma) \longrightarrow 0,$$

where  $A_{r-1}(X_\Sigma)$  is the Chow group of all divisors modulo linear equivalence.

**Remark 2.19.** The group  $A_{r-1}(X_\Sigma)$  is finitely generated and has the form

$$A_{r-1}(X_\Sigma) \cong \mathbb{Z}^s \oplus H,$$

where  $H$  is finite torsion. In particular,

$$A_{r-1}(X_\Sigma)/\text{torsion} \cong \mathbb{Z}^s.$$

**Proposition 2.20** (Gauge group from divisor classes). *Applying  $\text{Hom}(-, \mathbb{C}^*)$  to the exact sequence above yields*

$$0 \longrightarrow \text{Hom}(A_{r-1}(X_\Sigma), \mathbb{C}^*) \longrightarrow \text{Hom}(\mathbb{Z}^{\Sigma(1)}, \mathbb{C}^*) \longrightarrow \text{Hom}(M, \mathbb{C}^*) \longrightarrow 0.$$

Identifying  $\text{Hom}(\mathbb{Z}^{\Sigma(1)}, \mathbb{C}^*) \cong (\mathbb{C}^*)^{\Sigma(1)}$  and  $\text{Hom}(M, \mathbb{C}^*) \cong T$ , we obtain

$$G \cong \text{Hom}(A_{r-1}(X_\Sigma), \mathbb{C}^*).$$

If  $A_{r-1}(X_\Sigma)$  has no torsion, then  $G \cong (\mathbb{C}^*)^s$ .

**Definition 2.21** (Relation lattice). The lattice of relations among the toric divisors is

$$\Lambda := \text{Hom}(A_{r-1}(X_\Sigma), \mathbb{Z}) \cong H_2(X_\Sigma, \mathbb{Z}).$$

A basis  $\{C_a\}_{a=1}^s$  of  $\Lambda$  may be chosen to consist of homology classes of irreducible curves.

**Theorem 2.22** (Charges as intersection numbers). *Let  $Q_{i,a}$  be the GLSM charge matrix associated to  $X_\Sigma$ . Then*

$$Q_{i,a} = D_i \cdot C_a,$$

i.e. the charge of the  $i$ -th chiral superfield under the  $a$ -th  $U(1)$  is equal to the intersection number of the corresponding divisor  $D_i$  with the curve class  $C_a$ .

**Remark 2.23** (Meaning of the charge–intersection theorem). The content of Theorem 2.22 is not to introduce new geometric data, but to identify the GLSM charge matrix with the intersection pairing between divisors and curves.

More precisely, since the relation lattice

$$\Lambda := \text{Hom}(A_{r-1}(X_\Sigma), \mathbb{Z}) \cong H_2(X_\Sigma, \mathbb{Z})$$

has rank  $s$ , one may choose a basis  $\{C_a\}_{a=1}^s$  of  $H_2(X_\Sigma, \mathbb{Z})$  such that the intersection numbers with the toric divisors satisfy

$$Q_{i,a} = D_i \cdot C_a.$$

Thus the GLSM charge matrix  $Q$  is simply the matrix representation of the natural pairing

$$\text{Pic}(X_\Sigma) \times H_2(X_\Sigma, \mathbb{Z}) \longrightarrow \mathbb{Z}$$

in the chosen bases  $\{D_i\}$  and  $\{C_a\}$ .

In particular, the number of  $U(1)$  gauge factors in the GLSM equals the rank of  $H_2(X_\Sigma, \mathbb{Z})$ , and the choice of a basis of  $H_2$  corresponds to a choice of basis for the gauge group.

**Proposition 2.24** (Mori cone generators and curve basis). *Let  $X_\Sigma$  be a smooth projective toric variety of dimension  $r$ . Then the Mori cone  $\overline{\text{NE}}(X_\Sigma) \subset H_2(X_\Sigma, \mathbb{R})$  is generated by the classes of torus-invariant curves*

$$C_\tau := V(\tau), \quad \tau \in \Sigma(r-1),$$

*corresponding to  $(r-1)$ -dimensional cones of the fan.*

Moreover, since

$$\text{rank} H_2(X_\Sigma, \mathbb{Z}) = s = |\Sigma(1)| - r,$$

*one may choose a subset of  $s$  linearly independent such curve classes*

$$\{C_a\}_{a=1}^s \subset \{C_\tau\}_{\tau \in \Sigma(r-1)}$$

*which forms a  $\mathbb{Z}$ -basis of  $H_2(X_\Sigma, \mathbb{Z})$ .*

*Proof.* The first statement is a standard and nontrivial result in toric geometry which we choose not to prove it here: for smooth projective toric varieties, every effective one-cycle is numerically equivalent to a nonnegative linear combination of torus-invariant curves corresponding to  $(r-1)$ -dimensional cones.

Since these curves generate the Mori cone, their real span equals  $H_2(X_\Sigma, \mathbb{R})$ . Choosing  $s$  linearly independent generators then yields a  $\mathbb{Z}$ -basis of  $H_2(X_\Sigma, \mathbb{Z})$ .  $\square$

**Remark 2.25** (Intersection coordinates and the charge matrix). Fix a basis  $\{C_a\}_{a=1}^s$  of  $H_2(X_\Sigma, \mathbb{Z})$  consisting of torus-invariant curves as in the proposition. Then the intersection pairing induces an identification

$$A_{r-1}(X_\Sigma) \cong \text{Hom}(H_2(X_\Sigma, \mathbb{Z}), \mathbb{Z}),$$

and each divisor class  $D_i$  is represented by its intersection numbers

$$D_i \longleftrightarrow (D_i \cdot C_1, \dots, D_i \cdot C_s).$$

Under this identification, the GLSM charge matrix  $Q$  is precisely the matrix of the intersection pairing between toric divisors and the chosen curve basis:

$$Q_{i,a} = D_i \cdot C_a.$$

Thus the  $i$ -th row of  $Q$  gives the coordinates of the divisor  $D_i$  in the Chow group, while the choice of a basis of  $H_2(X_\Sigma, \mathbb{Z})$  corresponds to a choice of basis for the  $U(1)^s$  gauge group in the GLSM.

**Proposition 2.26** (FI parameters and the effective cone). *Let  $Q_{i,a}$  be the GLSM charge matrix and  $S = \{v_1, \dots, v_n\}$  the corresponding set of rays. Then the space of FI parameters is naturally identified with*

$$A_{n-1}(S) \otimes \mathbb{R} \cong \Lambda^* \otimes \mathbb{R},$$

where  $\Lambda \cong H_2(X_\Sigma, \mathbb{Z})$  is the relation lattice.

Moreover, the divisor classes of the torus-invariant divisors  $D_i$  span a cone

$$A_{n-1}^+(S) \otimes \mathbb{R} := \left\{ \sum_{i=1}^n x_i [D_i] \mid x_i \geq 0 \right\} \subset A_{n-1}(S) \otimes \mathbb{R},$$

called the effective divisor cone. The D-term equations admit a solution if and only if

$$r \in A_{n-1}^+(S) \otimes \mathbb{R}.$$

*Proof.* Set  $x_i := |\phi_i|^2 \in \mathbb{R}_{\geq 0}$ . Then the D-term equations are equivalent to the linear system

$$Q^T x = r, \quad x \in \mathbb{R}_{\geq 0}^n.$$

Thus a solution exists if and only if  $r$  lies in the image of the positive orthant under the linear map  $Q^T$ . By the identification

$$Q_{i,*} = [D_i] \in A_{n-1}(S),$$

the image of  $\mathbb{R}_{\geq 0}^n$  under  $Q^T$  is precisely the cone generated by the divisor classes  $[D_i]$ , namely the effective cone  $A_{n-1}^+(S) \otimes \mathbb{R}$ . Therefore the D-term equations admit a solution if and only if  $r \in A_{n-1}^+(S) \otimes \mathbb{R}$ .  $\square$

**Remark 2.27** (GKZ decomposition and GLSM phases). The effective cone  $A_{n-1}^+(S) \otimes \mathbb{R}$  admits a polyhedral decomposition into subcones, called the *GKZ (secondary) decomposition*. Each chamber corresponds to a choice of fan  $\Sigma'$  whose set of rays is  $S$ , and hence to a toric variety  $X_{\Sigma'}$ .

If the FI parameters  $r$  lie in the interior of a chamber, the space of supersymmetric ground states is a toric variety  $X_{\Sigma'}$ . Choosing  $r$  in different chambers produces different toric phases of the GLSM, related by wall-crossing transitions when  $r$  crosses the boundaries.

**Remark 2.28** (Kähler cone and the large volume phase). Among all GKZ chambers, the Kähler cone  $\mathcal{K}(X_\Sigma)$  of a fixed toric variety  $X_\Sigma$  selects the chamber corresponding to  $\Sigma$ . If the FI parameters lie in the interior of  $\mathcal{K}(X_\Sigma)$ , then the space of supersymmetric ground states is precisely  $X_\Sigma$ .

If the basis of  $\Lambda \cong H_2(X_\Sigma, \mathbb{Z})$  is chosen to generate the Mori cone, so that each basis vector is represented by an effective curve  $C_a$ , then the Kähler cone is the dual cone and the condition

$$r \in \mathcal{K}(X_\Sigma)$$

is simply

$$r_a > 0 \quad \text{for all } a.$$

This choice of basis corresponds to the physical “large volume phase” in which all FI parameters are positive.

**Example 2.29** (The total space of  $\mathcal{O}_{\mathbb{CP}^2}(-3)$ ). We illustrate the charge–intersection correspondence in a non-compact example by reconsidering the total space of the line bundle  $\mathcal{O}_{\mathbb{CP}^2}(-3)$ .

Let  $X_\Sigma$  be the toric variety associated with the fan of Example 7.2.5. The matrices  $P|Q$  describing the toric data are

$$P|Q = \left( \begin{array}{ccc|c} -1 & -1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -3 \end{array} \right),$$

where the four rows correspond respectively to the toric divisors

$$D_1 = \{x_1 = 0\}, \quad D_2 = \{x_2 = 0\}, \quad D_3 = \{x_3 = 0\}, \quad D_4 = \{\text{zero section}\}.$$

The unique column of  $Q$  represents a curve class

$$C \in H_2(X_\Sigma, \mathbb{Z}),$$

since  $\text{rk} H_2(X_\Sigma, \mathbb{Z}) = 1$  in this example. The GLSM charges are therefore  $(1, 1, 1, -3)$ , in agreement with the GLSM description of  $\mathcal{O}_{\mathbb{CP}^2}(-3)$  discussed earlier. Geometrically, the curve  $C$  is the zero section over a line in  $\mathbb{CP}^2$ . It intersects each of the divisors  $D_1, D_2, D_3$  transversely in a single point, hence

$$D_i \cdot C = 1, \quad i = 1, 2, 3.$$

Since  $C$  is contained in the zero section  $D_4$ , the intersection number  $C \cdot D_4$  is given by the degree of the normal bundle of  $D_4$  restricted to  $C$ :

$$C \cdot D_4 = \deg N_{D_4/X_\Sigma}|_C.$$

From the charge matrix we read off

$$C \cdot D_4 = -3,$$

which shows that the normal bundle of the zero section is  $\mathcal{O}_{\mathbb{CP}^2}(-3)$ . We thus recover the geometric interpretation

$$X_\Sigma \cong \text{Tot}(\mathcal{O}_{\mathbb{CP}^2}(-3)),$$

purely from the toric charge–intersection data.

## 3 Constructions in Toric Geometry

### 3.1 Orbifolds

**Definition 3.1** (Finite quotient singularities and orbifolds). Let  $X$  be an irreducible variety of dimension  $n$ .

- (a) A point  $p \in X$  is called a *finite quotient singularity* if there exists a finite subgroup  $G \subset \text{GL}(n, \mathbb{C})$  such that, in a neighborhood of  $p$ , the variety  $X$  is analytically isomorphic to the quotient  $\mathbb{C}^n/G$  near the origin.
- (b) The variety  $X$  is called an *orbifold* (or *quasismooth*, or said to have *finite quotient singularities*) if every point  $p \in X$  is a finite quotient singularity.

**Definition 3.2** (Simplicial cones and simplicial fans). A rational polyhedral cone  $\sigma \subset N_{\mathbb{R}}$  is called *simplicial* if it can be generated by linearly independent vectors  $v_1, \dots, v_k$ , i.e.

$$\sigma = \mathbb{R}_{\geq 0} \langle v_1, \dots, v_k \rangle \quad \text{with} \quad \dim \sigma = k.$$

Equivalently, the generators form a basis of the real vector space  $\text{Span}_{\mathbb{R}}(\sigma)$ .

A fan  $\Sigma$  is called *simplicial* if every cone  $\sigma \in \Sigma$  is simplicial.

**Proposition 3.3** (Orbifold criterion for toric varieties). *Let  $X_{\Sigma}$  be a toric variety of dimension  $r$ . Then  $X_{\Sigma}$  is an orbifold if and only if the fan  $\Sigma$  is simplicial.*

*Proof.* ( $\Leftarrow$ ) Let  $\sigma \in \Sigma$  be an  $r$ -dimensional cone generated by  $v_1, \dots, v_r$ . Since these vectors are linearly independent over  $\mathbb{R}$ , the associated affine toric variety

$$X_{\sigma} = \text{Spec } \mathbb{C}[\sigma^{\vee} \cap M]$$

is isomorphic to a quotient  $\mathbb{C}^r/G$  by a finite abelian group  $G$ . Hence  $X_{\sigma}$  has only finite quotient singularities and is an orbifold. Since  $X_{\Sigma}$  is covered by such affine charts, it follows that  $X_{\Sigma}$  is an orbifold.

( $\Rightarrow$ ) Assume now that  $X_{\Sigma}$  is a toric orbifold. Let  $\sigma \in \Sigma$  be an  $r$ -dimensional cone and let  $x = Z_{\sigma}$  be the corresponding torus-fixed point. Since  $X_{\Sigma}$  is an orbifold, the singularity at  $x$  is a finite quotient singularity.

A general result in the literature states that if a rationally smooth variety of dimension  $r$  admits a torus action with an isolated fixed point  $x$  and only finitely many  $T$ -invariant irreducible curves through  $x$ , then the number of such curves equals  $r$ . In the toric case, the  $T$ -invariant curves through  $x$  are in bijection with the codimension-1 faces of  $\sigma$ .

Hence  $\sigma$  has exactly  $r$  codimension-1 faces, which implies that  $\sigma$  is simplicial. Since this holds for every maximal cone, the fan  $\Sigma$  is simplicial.  $\square$

**Remark 3.4** (Local orbifold groups). For a simplicial cone  $\sigma = \mathbb{R}_{\geq 0} \langle v_1, \dots, v_r \rangle$ , the finite group  $G$  appearing in the local model  $X_{\sigma} \simeq \mathbb{C}^r/G$  is canonically given by

$$G = N / \langle v_1, \dots, v_r \rangle,$$

where  $\langle v_1, \dots, v_r \rangle \subset N$  is the sublattice generated by the primitive ray generators.

**Remark 3.5** (Global quotient description of simplicial toric orbifolds). Let  $\Sigma$  be a simplicial fan in  $N_{\mathbb{R}}$ , and suppose there exists a sublattice

$$N' \subset N$$

such that every top-dimensional cone  $\sigma \in \Sigma(r)$  is generated by a  $\mathbb{Z}$ -basis of  $N'$ .

Since  $N'_{\mathbb{R}} = N_{\mathbb{R}}$ , we may regard  $\Sigma$  as a fan in  $N'_{\mathbb{R}}$ , which defines a smooth toric variety

$$X_{\Sigma, N'}.$$

However, the acting torus changes from  $T = N \otimes \mathbb{C}^*$  to

$$T' = N' \otimes \mathbb{C}^*.$$

The inclusion  $N' \hookrightarrow N$  induces a finite morphism of tori

$$T' \longrightarrow T,$$

whose kernel is the finite abelian group  $N/N'$ .

Consequently, the original toric variety  $X_\Sigma = X_{\Sigma, N}$  is obtained as a global quotient

$$X_\Sigma \cong X_{\Sigma, N'} / (N/N').$$

In particular, any simplicial toric variety is a global orbifold quotient of a smooth toric variety by a finite group.

**Example 3.6** (The  $\mathbb{Z}_3$  orbifold of  $\mathbb{CP}^2$ ). We consider a concrete orbifold example which will later be used in the construction of the mirror of plane cubic curves.

Recall that the dense torus  $T = (\mathbb{C}^*)^2$  embeds into  $\mathbb{CP}^2$  by

$$(t_1, t_2) \longmapsto (1, t_1, t_2).$$

Let  $\mathbb{Z}_3 \subset (\mathbb{C}^*)^2$  be the subgroup generated by  $(\omega, \omega^2)$ , where

$$\omega = e^{2\pi i/3}.$$

This generator extends to an action on  $\mathbb{CP}^2$  by coordinatewise multiplication

$$(1, t_1, t_2) \longmapsto (1, \omega t_1, \omega^2 t_2).$$

We denote the quotient by  $\mathbb{CP}^2/\mathbb{Z}_3$ .

To describe this quotient torically, we must first understand the torus

$$T' := T/\mathbb{Z}_3.$$

The lattice of one-parameter subgroups of  $T'$  is strictly larger than that of  $T$ . Indeed, the map

$$t \longmapsto (1, t^{1/3}, t^{2/3})$$

defines a one-parameter subgroup of  $T'$  which does not lift to  $T$ . Hence the lattice

$$N' = N + \mathbb{Z}\left(\frac{1}{3}, \frac{2}{3}\right)$$

strictly contains  $N$ .

Since  $N'_\mathbb{R} = N_\mathbb{R}$ , we may take the same fan  $\Sigma$  as for  $\mathbb{CP}^2$ , but now regarded as a fan in the enlarged lattice  $N'$ . The resulting toric variety is denoted

$$X_{\Sigma, N'} = \mathbb{CP}^2/\mathbb{Z}_3.$$

Concretely, the generators of the one-dimensional cones of  $\Sigma$  are

$$(-1, -1), \quad (1, 0), \quad (0, 1).$$

If we change to coordinates adapted to a basis

$$\left\{ \left( \frac{2}{3}, \frac{1}{3} \right), \left( \frac{1}{3}, \frac{2}{3} \right) \right\}$$

of  $N'$ , then the same rays are represented by

$$(2, -1), \quad (-1, 2), \quad (-1, -1).$$

To see the orbifold singularities locally, consider the two-dimensional cone

$$\sigma = \text{Cone}((2, -1), (-1, 2)).$$

Since

$$\det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 3,$$

the vectors  $(2, -1)$  and  $(-1, 2)$  generate a sublattice of index 3 in  $N'$ . Therefore the corresponding affine toric variety is

$$X_\sigma \cong \mathbb{C}^2 / \mathbb{Z}_3.$$

This explicitly exhibits  $\mathbb{CP}^2 / \mathbb{Z}_3$  as a toric orbifold with local quotient singularities of type  $\mathbb{C}^2 / \mathbb{Z}_3$ .

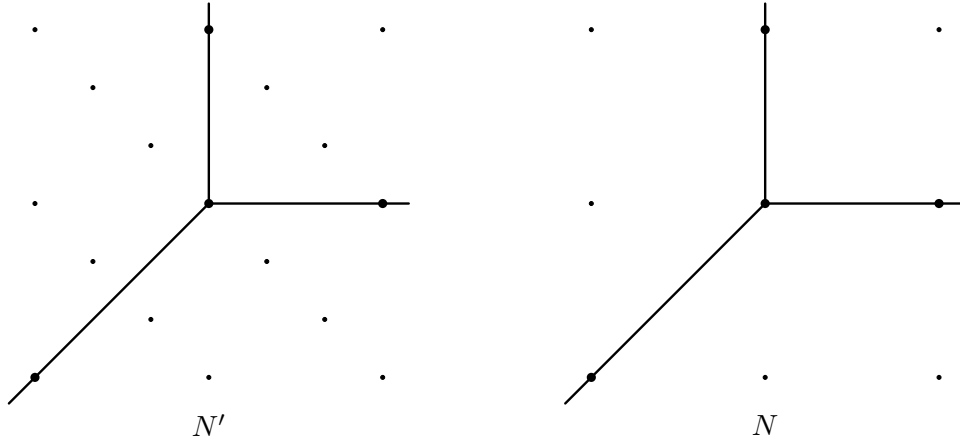


Figure 2: The fans for  $\mathbb{CP}^2 / \mathbb{Z}_3$  (lattice  $N'$ ) and  $\mathbb{CP}^2$  (lattice  $N$ ).

### 3.2 Blow-Up

**Definition 3.7** (Subdivision of a fan). Let  $\Sigma$  and  $\Sigma'$  be fans in the same lattice  $N$ . We say that  $\Sigma'$  *subdivides*  $\Sigma$  if the following conditions hold:

1.  $\Sigma(1) \subset \Sigma'(1)$ ;
2. every cone of  $\Sigma'$  is contained in some cone of  $\Sigma$ .

**Proposition 3.8** (Morphisms induced by fan subdivisions). *Let  $\Sigma'$  be a subdivision of a fan  $\Sigma$  in the same lattice  $N$ . Write*

$$\Sigma'(1) = \{\rho_1, \dots, \rho_m\}, \quad \Sigma(1) = \{\rho_1, \dots, \rho_n\} \subset \Sigma'(1),$$

where the rays are ordered so that the first  $n$  rays coincide.

Then the projection

$$\pi : \mathbb{C}^m \longrightarrow \mathbb{C}^n, \quad (x_1, \dots, x_m) \longmapsto (x_1, \dots, x_n)$$

induces a well-defined toric morphism

$$f : X_{\Sigma'} \longrightarrow X_{\Sigma}.$$

Moreover,  $f$  is birational and restricts to the identity on the dense torus  $T$ .

*Proof.* Recall that

$$X_{\Sigma'} = (\mathbb{C}^m \setminus Z(\Sigma'))/G_{\Sigma'}, \quad X_{\Sigma} = (\mathbb{C}^n \setminus Z(\Sigma))/G_{\Sigma}.$$

Step 1: Compatibility with the irrelevant loci. Since  $\Sigma'$  subdivides  $\Sigma$ , every cone of  $\Sigma'$  is contained in a cone of  $\Sigma$ . This implies that a set of coordinates vanishing corresponds to a forbidden set for  $\Sigma'$  only if the same set is forbidden for  $\Sigma$ . Hence the projection  $\pi$  satisfies

$$\pi(\mathbb{C}^m \setminus Z(\Sigma')) \subset \mathbb{C}^n \setminus Z(\Sigma).$$

Step 2: Compatibility with the group actions. The inclusion of rays  $\Sigma(1) \subset \Sigma'(1)$  induces a natural homomorphism

$$G_{\Sigma'} \longrightarrow G_{\Sigma},$$

with respect to which the projection  $\pi$  is equivariant. Therefore  $\pi$  descends to a morphism

$$f : X_{\Sigma'} \longrightarrow X_{\Sigma}.$$

Step 3: Birationality. Both  $X_{\Sigma'}$  and  $X_{\Sigma}$  contain the same dense torus  $T$ , and the morphism  $f$  restricts to the identity on  $T$ . Hence  $f$  is an isomorphism on a dense open subset and therefore birational.  $\square$

**Remark 3.9.** It should be noted that the construction described above is intentionally simplified. However, this simplification may cause confusion, since the naive projection does not automatically yield a well-defined map on the quotient without taking the group action into account. To obtain a morphism on the quotient, one must first choose representatives in each  $G$ -orbit, we will discuss it in more detail later.

**Proposition 3.10** (Toric blow-up as star subdivision). *Let  $X_{\Sigma}$  be a smooth projective toric variety of dimension  $r$ , and let  $p \in X_{\Sigma}$  be a  $T$ -invariant smooth point. Let  $\sigma \in \Sigma$  be the  $r$ -dimensional cone corresponding to  $p$ , with primitive generators*

$$\sigma = \langle v_1, \dots, v_r \rangle \subset N_{\mathbb{R}}.$$

*Define a new ray*

$$\rho_{r+1} := \mathbb{R}_{\geq 0} v_{r+1}, \quad v_{r+1} := v_1 + \dots + v_r \in N.$$

*Let  $\Sigma'$  be the fan obtained from  $\Sigma$  by replacing the cone  $\sigma$  with the  $r$  cones*

$$\langle v_1, \dots, \widehat{v_i}, \dots, v_r, v_{r+1} \rangle, \quad i = 1, \dots, r,$$

*and keeping all other cones of  $\Sigma$  unchanged. Then  $X_{\Sigma'}$  is the blow-up of  $X_{\Sigma}$  at the point  $p$ .*

*Proof.* The cone  $\sigma$  corresponds to the  $T$ -fixed point

$$Z_\sigma = V(\sigma) \subset X_\Sigma,$$

which is a smooth point, so  $\sigma$  is generated by a  $\mathbb{Z}$ -basis of  $N$ . The star subdivision replaces the affine toric chart

$$U_\sigma \simeq \mathbb{C}^r$$

by the toric variety associated to  $\Sigma'|_\sigma$ , which is exactly the toric model of the blow-up of  $\mathbb{C}^r$  at the origin.

Gluing with the unchanged cones of  $\Sigma$  yields a birational morphism

$$\pi: X_{\Sigma'} \longrightarrow X_\Sigma$$

which is an isomorphism away from  $Z_\sigma$ , and hence realizes the blow-up of  $X_\Sigma$  at the point  $p = Z_\sigma$ .  $\square$

**Remark 3.11.** The following example illustrates concretely the key point behind the construction of the toric morphism: what matters is the existence of a projection that is well defined on the quotient by the group action.

Consider the toric blow-up of  $\mathbb{C}^n$  at the origin. In Cox coordinates it can be written as

$$X_{\Sigma'} = (\mathbb{C}^{n+1} \setminus Z)/G, \quad G = \{(t^{-1}, t, t, \dots, t) \mid t \in \mathbb{C}^*\}.$$

If one attempts to use the naive projection

$$\pi_{\text{naive}} : (x_0, x_1, \dots, x_n) \longmapsto (x_1, \dots, x_n),$$

then this map is not compatible with the  $G$ -action. Indeed, for  $g = (t^{-1}, t, \dots, t) \in G$  we have

$$\pi_{\text{naive}}(g \cdot (x_0, x_1, \dots, x_n)) = (tx_1, \dots, tx_n) \neq \pi_{\text{naive}}(x_0, x_1, \dots, x_n),$$

so  $\pi_{\text{naive}}$  does not descend to a map on the quotient.

To obtain a morphism on the quotient, one must first choose representatives in each  $G$ -orbit (i.e. take a slice compatible with the group action). For instance, when  $x_0 \neq 0$ , every orbit contains a representative of the form

$$(1, x_0x_1, x_0x_2, \dots, x_0x_n).$$

On such representatives we define the modified projection

$$\pi(x_0, x_1, \dots, x_n) = (x_0x_1, x_0x_2, \dots, x_0x_n).$$

This map satisfies

$$\pi(g \cdot x) = \pi(x) \quad (\forall g \in G),$$

and hence is invariant under the  $G$ -action, so it induces a well-defined map on the quotient.

In particular, when  $x_0 = 0$ , the above formula gives  $\pi(x) = (0, \dots, 0)$ , which corresponds to points on the exceptional divisor mapping to the origin. Thus the apparent paradox that “the origin has no preimage” arises only from using a projection that is not equivariant; once an equivariant representative is chosen, the projection becomes well defined and the exceptional divisor correctly appears as the fiber over the origin.

This shows that, in constructing toric morphisms via Cox coordinates, the essential step is to define the projection on representatives compatible with the group action, rather than using the naive coordinate projection on the ambient affine space.

**Remark 3.12** (Exceptional divisor). The new ray  $\rho_{r+1}$  corresponds to a torus-invariant divisor

$$E := D_{\rho_{r+1}} \subset X_{\Sigma'}.$$

This divisor is the *exceptional divisor* of the blow-up. More precisely,

$$E = \pi^{-1}(Z_\sigma),$$

and  $E \simeq \mathbb{P}^{r-1}$ , with normal bundle

$$\mathcal{N}_{E/X_{\Sigma'}} \simeq \mathcal{O}_{\mathbb{P}^{r-1}}(-1).$$

The point  $Z_\sigma$  itself is *not* the exceptional locus; rather, it is replaced by the divisor  $E$  in the blow-up.

**Remark 3.13** (Blow-up of an ideal). More generally, if  $\Sigma'$  is obtained from  $\Sigma$  by an arbitrary subdivision, the induced morphism  $X_{\Sigma'} \rightarrow X_\Sigma$  is the blow-up of a  $T$ -invariant ideal sheaf  $\mathcal{I} \subset \mathcal{O}_{X_\Sigma}$ . This ideal is supported on the union of all torus-invariant subvarieties  $V(\tau)$  such that  $\tau \in \Sigma$  but  $\tau \notin \Sigma'$ .

In the special case before, the only cone of  $\Sigma$  that is not a cone of  $\Sigma'$  is  $\sigma$ , hence the center of the blow-up is exactly the point  $Z_\sigma$ .

**Remark 3.14** (Comparison with Algebraic Geometry). In the usual language, the blow-up of a smooth variety  $X$  at a point  $p \in X$  is defined as

$$\mathrm{Bl}_p X := \mathrm{Proj} \bigoplus_{n \geq 0} \mathfrak{m}_p^n,$$

where  $\mathfrak{m}_p$  is the maximal ideal of  $p$ . The exceptional divisor is

$$E \simeq \mathbb{P}(T_p X) \simeq \mathbb{P}^{r-1}.$$

The toric construction realizes this abstract definition combinatorially.

**Example 3.15** (Resolution of the  $A_1$  singularity). We consider the toric resolution of the orbifold  $\mathbb{C}^2/\{\pm 1\}$  at the origin. We choose a lattice  $N$  such that the fan of  $\mathbb{C}^2/\{\pm 1\}$  consists of the cone

$$\sigma = \mathrm{Cone}(v_1, v_2), \quad v_1 = (1, 0), \quad v_2 = (1, 2),$$

together with its faces. This cone has determinant 2 and hence corresponds to an  $A_1$  surface singularity.

To resolve this singularity, we insert the ray generated by

$$v_3 = (1, 1),$$

and subdivide  $\sigma$  into two cones

$$\mathrm{Cone}(v_1, v_3) \quad \text{and} \quad \mathrm{Cone}(v_3, v_2).$$

Both cones are basic (their generators form a  $\mathbb{Z}$ -basis of  $N$ ), so the resulting fan  $\Sigma$  defines a smooth toric surface  $X_\Sigma$ . This toric variety is the minimal resolution of  $\mathbb{C}^2/\{\pm 1\}$  at the singular point.

The new ray  $\rho_3 = \text{Cone}(v_3)$  corresponds to a torus-invariant divisor  $D_3$ , which is the exceptional divisor of the resolution. The lattice relation

$$v_1 + v_2 - 2v_3 = 0$$

gives the charge vector  $(1, 1, -2)$ . Equivalently, using the standard toric intersection formula for smooth toric surfaces,

$$v_1 + v_2 = 2v_3 \implies D_3^2 = -2.$$

Thus the exceptional divisor is a  $(-2)$ -curve, in agreement with the classical resolution of an  $A_1$  singularity.

**Remark 3.16.** The above computation requires some care in interpretation. Although the Mori cone of  $X_\Sigma$  is one-dimensional, its generators must be represented by *proper* (i.e. compact) curves. Among the torus-invariant divisors  $D_1, D_2, D_3$ , only  $D_3$  is compact; the divisors  $D_1$  and  $D_2$  are non-compact and therefore do not represent classes in the Mori cone.

From a cohomological viewpoint, intersection numbers are defined via Poincaré duality, which pairs divisor classes with homology classes represented by compact cycles. Hence, when working on a non-compact toric surface, one is naturally restricted to compact curve classes in  $H_2$ . This explains why the Mori cone is generated by the class of  $D_3$  and not by arbitrary combinations of  $D_1$  and  $D_2$ .

Consequently, the self-intersection number  $D_3^2 = -2$  is intrinsic: it is determined solely by the local toric geometry through the relation  $v_1 + v_2 = 2v_3$ , and not by an arbitrary choice of a generator of  $H_2$ . The other divisors do not play a symmetric role, since they do not correspond to proper curves and therefore do not define intersection classes in the same sense.

**Remark 3.17.** The previous example extends naturally to the resolution of an  $A_n$  singularity. Recall that an  $A_n$  surface singularity can be written as the quotient

$$\mathbb{C}^2 / \mathbb{Z}_{n+1},$$

where the generator of  $\mathbb{Z}_{n+1}$  acts on  $\mathbb{C}^2$  by

$$(z_1, z_2) \longmapsto (\omega z_1, \omega^n z_2), \quad \omega = \exp(2\pi i / (n+1)).$$

A toric model for this singularity is given by the cone

$$\sigma = \text{Cone}((1, 0), (1, n+1)) \subset N_{\mathbb{R}}.$$

To obtain a resolution, we subdivide  $\sigma$  by inserting the rays generated by

$$v_i = (1, i), \quad i = 1, \dots, n.$$

The resulting fan  $\Sigma'$  defines a smooth toric surface  $X_{\Sigma'}$ .

For each interior ray  $v_i$  with  $1 \leq i \leq n$ , the lattice relations

$$v_{i-1} + v_{i+1} - 2v_i = 0$$

imply, as in the  $A_1$  case, that the corresponding torus-invariant divisors satisfy

$$D_i^2 = -2.$$

These divisors are all compact and represent the generators of the Mori cone. Moreover, they intersect according to the pattern

$$D_i \cdot D_{i+1} = 1, \quad D_i \cdot D_j = 0 \text{ for } |i - j| > 1,$$

so that the divisors  $D_1, \dots, D_n$  form a chain of rational curves. This recovers the classical minimal resolution of an  $A_n$  singularity as a chain of  $n$   $(-2)$ -curves.

**Example 3.18** (A non-simplicial cone and the conifold/node). Let  $N \cong \mathbb{Z}^3$  and consider the fan  $\Sigma$  consisting of the cone

$$\sigma = \text{Cone}(v_1, v_2, v_3, v_4) \subset N_{\mathbb{R}}, \quad v_1 = (1, 0, 0), \quad v_2 = (0, 1, 0), \quad v_3 = (0, 0, 1), \quad v_4 = (1, 1, -1),$$

together with all of its faces. Then  $X_{\Sigma} = U_{\sigma}$  is singular.

Since  $\dim \sigma = 3$  but  $\sigma$  is generated by 4 rays, the cone  $\sigma$  is not simplicial. Equivalently,  $\sigma$  cannot be written as  $\text{Cone}(w_1, w_2, w_3)$  with  $w_i$  linearly independent in  $N_{\mathbb{R}}$ . Consequently, the singularity of  $X_{\Sigma}$  is not an orbifold (quotient) singularity.

This singularity is called a *node* in the mathematics literature and a *conifold* singularity in the physics literature. It admits two distinct toric blow-ups (small resolutions) producing smooth toric varieties, obtained by subdividing  $\sigma$  without introducing any new rays. In particular, there is no exceptional divisor. In each subdivision there appears a new two-dimensional cone  $\tau$  (namely  $\tau = \text{Cone}(v_1, v_2)$  in one case and  $\tau = \text{Cone}(v_3, v_4)$  in the other), hence an exceptional curve  $Z_{\tau} \subset X_{\Sigma'}$ . This curve is a compact one-dimensional toric variety and therefore isomorphic to  $\mathbb{CP}^1$ . The induced birational map between the two small resolutions is a *flop*.

**Theorem 3.19** (Toric resolution of singularities). *For any fan  $\Sigma$  in a lattice  $N$ , there exists a refinement  $\tilde{\Sigma}$  such that the induced toric morphism*

$$X_{\tilde{\Sigma}} \longrightarrow X_{\Sigma}$$

*is a resolution of singularities. That is,  $X_{\tilde{\Sigma}}$  is smooth, and the morphism is proper, birational, and an isomorphism over the smooth locus of  $X_{\Sigma}$ .*

*Proof.* We only present the idea of the proof. For a cone  $\sigma \subset N_{\mathbb{R}}$ , the corresponding affine toric variety  $U_{\sigma}$  is smooth if and only if  $\sigma$  is generated by a part of a  $\mathbb{Z}$ -basis of  $N$ . More generally,  $U_{\sigma}$  has only orbifold (quotient) singularities if  $\sigma$  is simplicial. Starting from an arbitrary fan  $\Sigma$ , one can subdivide each non-smooth cone by inserting additional rays generated by primitive lattice points. A classical algorithm is to take barycentric subdivisions or stellar subdivisions along carefully chosen lattice vectors. After finitely many steps, every maximal cone becomes basic, producing a smooth fan  $\tilde{\Sigma}$ . The morphism  $X_{\tilde{\Sigma}} \rightarrow X_{\Sigma}$  induced by the refinement is proper, birational, and an isomorphism over the smooth locus, hence a resolution of singularities.  $\square$

**Remark 3.20** (Comments on toric resolution of singularities). Theorem 3.19 states that every toric variety admits a toric resolution of singularities. This result is fundamentally combinatorial: singularities of a toric variety are encoded entirely in the structure of its fan, and resolving them amounts to refining the fan so that all cones become smooth (or at least simplicial).

Unlike Hironaka's resolution of singularities for general algebraic varieties, which relies on deep analytic and algebro-geometric techniques, the toric case is completely explicit and constructive. The existence of  $\tilde{\Sigma}$  can be viewed as a toric analogue of desingularization by successive blow-ups, but carried out purely at the level of polyhedral geometry.

### 3.3 Morphisms

**Definition 3.21.** Let  $\Sigma$  be a fan in  $N_{\mathbb{R}}$  and let  $\Sigma'$  be a fan in  $N'_{\mathbb{R}}$ . A morphism from  $\Sigma$  to  $\Sigma'$  consists of a homomorphism  $\psi : N \rightarrow N'$  such that for each  $\sigma \in \Sigma$ , the image of  $\sigma$  under  $\psi \otimes \mathbb{R}$  is contained in some cone of  $\Sigma'$ .

The mapping  $\psi : N \rightarrow N'$  induces a natural mapping of tori

$$T = N \otimes \mathbb{C} \longrightarrow T' = N' \otimes \mathbb{C}.$$

It is no worth checking that this extends to a mapping  $X_{\Sigma} \rightarrow X_{\Sigma'}$ . The global orbifold considered before gives a class of simple examples. In that case, we have  $N'$  is a sublattice of  $N$ ,  $\psi : N' \rightarrow N$  is the inclusion mapping,  $\psi \otimes \mathbb{R}$  is the identity map, and  $\Sigma = \Sigma'$ .

**Example 3.22.** Consider the toric variety  $X = \mathbb{CP}^1 \times \mathbb{CP}^1$ . Its fan  $\Sigma \subset \mathbb{R}^2$  has four one-dimensional cones generated by

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad v_3 = (-1, 0), \quad v_4 = (0, -1).$$

These rays determine four  $T$ -invariant divisors and correspondingly four  $T$ -invariant curves on  $X$ .

There are two natural lattice homomorphisms

$$\pi_1 : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \quad (x, y) \mapsto x,$$

$$\pi_2 : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \quad (x, y) \mapsto y,$$

each given by projection onto one of the coordinates.

Both  $\pi_1$  and  $\pi_2$  define morphisms of fans  $\Sigma \rightarrow \Sigma_{\mathbb{CP}^1}$ , where  $\Sigma_{\mathbb{CP}^1}$  is the standard fan of  $\mathbb{CP}^1$  with rays generated by 1 and  $-1$ .

Indeed, for every cone  $\sigma \in \Sigma$ , its image under either projection is contained in one of the cones of  $\Sigma_{\mathbb{CP}^1}$ . Therefore, each projection induces a morphism of toric varieties

$$X_{\Sigma} = \mathbb{CP}^1 \times \mathbb{CP}^1 \longrightarrow \mathbb{CP}^1.$$

These are precisely the two coordinate projections onto the first and second factors, respectively.

**Example 3.23.** Let  $n > 0$  and consider the toric surface  $F_n$ , whose fan  $\Sigma_{F_n} \subset \mathbb{R}^2$  is generated by the rays

$$(1, 0), \quad (0, 1), \quad (-1, n), \quad (0, -1).$$

Consider the two natural lattice homomorphisms

$$\pi_1 : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \quad (x, y) \mapsto x,$$

$$\pi_2 : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \quad (x, y) \mapsto y.$$

(1) Projection onto the first factor.

The map  $\pi_1$  defines a morphism of fans

$$\Sigma_{F_n} \longrightarrow \Sigma_{\mathbb{CP}^1},$$

since for every cone  $\sigma \in \Sigma_{F_n}$ , the image  $(\pi_1 \otimes \mathbb{R})(\sigma)$  is contained in a cone of the standard fan of  $\mathbb{CP}^1$ . Hence  $\pi_1$  induces a morphism of toric varieties

$$F_n \longrightarrow \mathbb{CP}^1.$$

This realizes  $F_n$  as a  $\mathbb{CP}^1$ -bundle over  $\mathbb{CP}^1$ .

(2) Projection onto the second factor.

The map  $\pi_2$ , however, does *not* define a morphism of fans. Indeed, consider the cone in  $\Sigma_{F_n}$  generated by the rays

$$(-1, n) \quad \text{and} \quad (0, 1).$$

Under  $\pi_2$ , these vectors map to  $n$  and  $1$ , which generate all of  $\mathbb{R}$ . Thus the image of this cone under  $\pi_2 \otimes \mathbb{R}$  is the entire line  $\mathbb{R}$ , which is not contained in any cone of the fan of  $\mathbb{CP}^1$ . Therefore  $\pi_2$  is not a morphism of fans, and no toric morphism

$$F_n \longrightarrow \mathbb{CP}^1$$

arises from this projection.

This reflects the fact that  $F_n$  is a nontrivial  $\mathbb{CP}^1$ -bundle over  $\mathbb{CP}^1$ .

(3) Local triviality.

From the toric viewpoint, the bundle is nevertheless locally trivial. Consider the affine open subset  $U \subset \mathbb{CP}^1$  obtained by removing the ray generated by  $(-1)$  from the fan of  $\mathbb{CP}^1$ ; this corresponds to  $U \simeq \mathbb{C}$ .

To restrict the bundle over this open set, we remove the corresponding ray  $(-1, n)$  from the fan of  $F_n$ . The resulting fan is easily seen to be the product fan of

$$\mathbb{C} \times \mathbb{CP}^1.$$

Performing the same analysis on the other affine chart of  $\mathbb{CP}^1$  shows that the bundle is locally a product over both standard affine opens. Hence  $F_n$  is a locally trivial  $\mathbb{CP}^1$ -bundle over  $\mathbb{CP}^1$ , although it is globally nontrivial.

**Example 3.24.** We return to the fan  $\Sigma$  of the total space  $\text{Tot}(\mathcal{O}_{\mathbb{CP}^2}(-3))$ , and observe that it can be constructed directly from the fan  $\Sigma'$  of  $\mathbb{CP}^2$ .

Each three-dimensional cone of  $\Sigma$  is spanned by

$$(v_1, 1), \quad (v_2, 1), \quad (0, 0, 1),$$

where  $\{v_1, v_2\}$  is any pair chosen from the set

$$\{(-1, -1), (1, 0), (0, 1)\}$$

that spans a two-dimensional cone of  $\Sigma'$ .

Projection onto the first two coordinates defines a lattice homomorphism

$$\pi : \mathbb{Z}^3 \longrightarrow \mathbb{Z}^2, \quad (x_1, x_2, x_3) \longmapsto (x_1, x_2),$$

which induces a morphism of fans  $\Sigma \rightarrow \Sigma'$ . Consequently, we obtain a toric morphism

$$\pi : \text{Tot}(\mathcal{O}_{\mathbb{CP}^2}(-3)) \longrightarrow \mathbb{CP}^2.$$

This morphism realizes the total space as a locally trivial line bundle over  $\mathbb{CP}^2$ . The global structure of this bundle, namely that it is  $\mathcal{O}_{\mathbb{CP}^2}(-3)$ , can be read directly from the toric data.

In general, if the edges of  $\Sigma'$  generated by  $v_i \in N$  are lifted to vectors

$$(v_i, k_i) \in N \oplus \mathbb{Z},$$

and an additional ray  $(0, \dots, 0, 1)$  is added, then the resulting toric variety is the total space of the line bundle

$$\mathcal{O}\left(-\sum k_i D_i\right),$$

where  $D_i$  are the torus-invariant divisors corresponding to the rays  $v_i$ .

In the present example, all  $k_i = 1$ , so the resulting bundle is

$$\mathcal{O}_{\mathbb{CP}^2}(-D_1 - D_2 - D_3) = \mathcal{O}_{\mathbb{CP}^2}(-3).$$

Note that the ray generated by  $(0, 0, 1)$  projects to the zero cone of  $\Sigma'$ . Since the toric subvariety corresponding to the zero cone is the whole base  $\mathbb{CP}^2$ , we conclude that the divisor  $D_{(0,0,1)}$  maps surjectively to  $\mathbb{CP}^2$ . Indeed, this divisor is precisely the zero section of the line bundle.

The remaining  $T$ -invariant divisors project to divisors in  $\mathbb{CP}^2$ :

$$D_{(-1,-1,1)} = \pi^{-1}(\{x_1 = 0\}), \quad D_{(1,0,1)} = \pi^{-1}(\{x_2 = 0\}), \quad D_{(0,1,1)} = \pi^{-1}(\{x_3 = 0\}).$$

**Example 3.25.** The discussion above can be generalized from line bundles to projective bundles, and even to weighted projective bundles. We now describe an example arising naturally in string theory.

Consider a fan  $\Sigma \subset \mathbb{Z}^4$  whose rays are given by the columns of the matrix

$$P = \begin{pmatrix} -1 & -2 & -2 & -3 \\ 1 & 0 & -2 & -3 \\ 0 & 1 & -2 & -3 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -2 & -3 \end{pmatrix}.$$

We leave it as an exercise to determine explicitly the cones of this fan.

Projection onto the first two coordinates defines a lattice map

$$\varphi : \mathbb{Z}^4 \longrightarrow \mathbb{Z}^2, \quad (x_1, x_2, x_3, x_4) \longmapsto (x_1, x_2),$$

which sends the fan  $\Sigma$  to the fan of the Hirzebruch surface  $F_2$ . Hence  $\varphi$  induces a toric morphism

$$X_\Sigma \longrightarrow F_2.$$

The fibers of this morphism are weighted projective spaces  $\mathbb{CP}(1, 2, 3)$ . Indeed, the rays of  $\Sigma$  that lie in the kernel of  $\varphi$  are

$$(0, 0, 1, 0), \quad (0, 0, 0, 1), \quad (0, 0, -2, -3).$$

Restricting to the sublattice  $\text{Ker}(\varphi) \cong \mathbb{Z}^2$ , these give the rays

$$(1, 0), \quad (0, 1), \quad (-2, -3),$$

which form the fan of the weighted projective space  $\mathbb{CP}(1, 2, 3)$ . They span a cone that vanishes after the projection, so the toric variety they spanned also vanishes. Thus this is just the fiber.

This toric variety contains Calabi–Yau hypersurfaces whose fibers over  $F_2$  are elliptic curves embedded in  $\mathbb{CP}(1, 2, 3)$ . Such constructions provide a standard method for producing elliptic fibrations, and play an important role in F-theory compactifications.

### 3.4 Local Geometry in Geometric Engineering

This subsection serves as a very brief introduction to the geometric objects we meet in geometric engineering.

The idea of geometric engineering is to construct geometric models with desired properties so that the resulting string theory, M-theory, or F-theory compactification has the desired physics. Toric geometry provides a useful way to engineer these geometries, as it is easy to perform direct physical computations in this framework.

For example, one way to produce an  $\mathcal{N} = 2$   $SU(2)$  gauge theory in four dimensions is to construct a Calabi–Yau threefold  $X$  containing a surface  $F_n$ , which can be blown down to the base  $\mathbb{CP}^1$ . We consider type IIA string theory compactified on  $X$ . There are two massive states corresponding to D2-branes wrapping the fibers of  $F_n$ , with either orientation. In the limit where the fiber shrinks and the base  $\mathbb{CP}^1$  becomes large in such a way as to decouple gravity, we obtain a field theory in which these massive states become massless and join with an existing  $U(1)$  associated to the volume of the fiber to form an  $SU(2)$  vector multiplet.

A local model for this geometry is the canonical bundle of  $F_n$ , with  $F_n$  embedded as the zero section. This model can be constructed explicitly using toric geometry.

To construct such local Calabi–Yau models, recall Example 7.7.4, where we showed how to build the fan of  $\mathcal{O}(K_{\mathbb{CP}^2}) = \mathcal{O}_{\mathbb{CP}^2}(-3)$  from the fan of  $\mathbb{CP}^2$ , and how this method can be generalized to more general bundles over toric varieties. In particular, we now apply this method to construct the canonical bundle over  $F_2$ .

The result can be described in terms of the matrices  $P|Q$  for  $F_2$  and for its canonical bundle  $K_{F_2}$ :

$$F_2 = \left( \begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ -1 & -2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & -2 \end{array} \right), \quad K_{F_2} = \left( \begin{array}{ccc|cc} 1 & 0 & 1 & 0 & 1 \\ -1 & -2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 1 & -2 \\ 0 & 0 & 1 & -2 & 0 \end{array} \right).$$

Once this toric data is available, we can directly derive field theory results using mirror symmetry. In particular, the prepotential can be obtained from an appropriate Picard–Fuchs system of differential equations, which can be deduced from the charge matrix  $Q$ .

We can also analyze this local geometry directly from the toric data. Projection onto the first coordinate defines a morphism

$$K_{F_2} \longrightarrow \mathbb{CP}^1,$$

for which the divisors  $D_1$  and  $D_2$  are fibers.

To understand the geometry of these divisors, we compute their associated fans as before. Using  $D_1$  as an example, we project the lattice onto

$$\mathbb{Z}^3 / \mathbb{Z} \cdot (1, 0, 1).$$

We use the explicit isomorphism

$$\mathbb{Z}^3 / \mathbb{Z} \cdot (1, 0, 1) \cong \mathbb{Z}^2, \quad [(a, b, c)] \longmapsto (c - a, b + c - a).$$

Under this identification, the divisor  $D_1$  is seen to be the toric variety associated with a two-dimensional fan whose rays are generated by

$$(1, 0), \quad (1, 1), \quad (1, 2).$$

This fan describes the minimal resolution of an  $A_1$  surface singularity. Consequently, the local Calabi–Yau threefold can be viewed as a resolution of an  $A_1$  singularity fibered over  $\mathbb{CP}^1$ .

More generally, this construction can be extended to produce fibrations of resolutions of  $A_n$  singularities over  $\mathbb{CP}^1$ . The Picard–Fuchs equations for the mirror geometry correspond to the Picard–Fuchs equations governing quantum cohomology, and these relations can be verified directly in many situations.

### 3.5 Polytopes

We now discuss projective toric varieties and their relationship with polytopes. Our polytopes will be in  $M_{\mathbb{R}}$ , the dual space of  $N_{\mathbb{R}}$ .

**Definition 3.26.** An integral polytope in  $M_{\mathbb{R}}$  is the convex hull of a finite set of points in  $M$ .

In the sequel, we will drop the adjective “integral” and refer to these simply as polytopes.

The  $r$ -dimensional polytopes are the data needed to describe projective toric varieties.

**Definition 3.27** (Toric variety from a polytope). Let  $\Delta \subset M_{\mathbb{R}}$  be an  $r$ -dimensional integral polytope. Choose an ordering of the lattice points

$$\Delta \cap M = \{m_0, m_1, \dots, m_k\}.$$

Since  $M = \text{Hom}(T, \mathbb{C}^*)$ , each  $m_i$  can be regarded as a nowhere-vanishing holomorphic function on the torus  $T$ . These functions determine a map

$$f : T \longrightarrow \mathbb{CP}^k, \quad f(t) = (m_0(t), \dots, m_k(t)).$$

We define  $\mathbb{CP}_{\Delta}$  to be the closure of  $f(T)$  in  $\mathbb{CP}^k$ . With the natural coordinatewise action of  $T$  on  $\mathbb{CP}^k$ , the variety  $\mathbb{CP}_{\Delta}$  acquires the structure of a projective toric variety. This construction is independent of the chosen ordering of  $\Delta \cap M$ .

**Remark 3.28.** If we write the homogeneous coordinates of  $\mathbb{CP}^k$  as  $(y_0, \dots, y_k)$ , then the image of  $f$  is described by the relations among the lattice points  $m_i$ . More precisely, if

$$\sum_i a_i m_i = 0 \quad \text{in } M \quad \text{with} \quad \sum_i a_i = 0,$$

then the image satisfies the homogeneous equation

$$\prod_{a_i > 0} y_i^{a_i} = \prod_{a_i < 0} y_i^{-a_i}.$$

Thus  $\mathbb{CP}_\Delta \subset \mathbb{CP}^k$  can be defined directly by such binomial equations.

**Remark 3.29.** If one uses only a subset of  $\Delta \cap M$  whose convex hull is still  $\Delta$ , the same construction produces a possibly non-normal toric variety. For example, taking  $\Delta = [0, k]$  but using only the lattice points  $\{0, a_1, \dots, a_\ell, k\}$  yields a non-normal curve of degree  $k$  in  $\mathbb{CP}^{\ell+1}$ .

**Definition 3.30** (Normal fan of a polytope). Assume now that we restrict to normal toric varieties. For each face  $F$  of the polytope  $\Delta$ , define the cone

$$\sigma_F = \{v \in N_{\mathbb{R}} \mid \langle m, v \rangle \leq \langle m', v \rangle \text{ for all } m \in F, m' \in \Delta\}.$$

The collection of cones  $\{\sigma_F\}$ , as  $F$  ranges over all faces of  $\Delta$ , forms a fan in  $N_{\mathbb{R}}$ , called the *normal fan* of  $\Delta$ , and is denoted by  $\Sigma_\Delta$ .

**Theorem 3.31.** Let  $\Delta \subset M_{\mathbb{R}}$  be an integral polytope and let  $\Sigma_\Delta$  be its normal fan. Then the toric variety associated to this fan satisfies

$$X_{\Sigma_\Delta} \simeq \mathbb{CP}_\Delta.$$

**Remark 3.32.** The isomorphism in the theorem can be described explicitly as follows. If  $\Sigma_\Delta(1) = \{v_1, \dots, v_n\}$  are the rays of the normal fan and  $(x_1, \dots, x_n)$  are the corresponding homogeneous coordinates on  $X_{\Sigma_\Delta}$ , then the map

$$(x_1, \dots, x_n) \mapsto \left( \prod_{i=1}^n x_i^{\langle m_0, v_i \rangle}, \dots, \prod_{i=1}^n x_i^{\langle m_k, v_i \rangle} \right)$$

realizes the identification  $X_{\Sigma_\Delta} \cong \mathbb{CP}_\Delta$ .

**Example 3.33.** We illustrate how projective toric varieties arise from lattice polytopes.

(1) *The standard simplex and  $\mathbb{CP}^2$ .*

Let

$$\Delta = \text{Conv}\{(0, 0), (1, 0), (0, 1)\} \subset \mathbb{R}^2.$$

Then

$$\Delta \cap M = \{(0, 0), (1, 0), (0, 1)\}.$$

By the construction of toric varieties from polytopes, the associated map from the torus  $T = (\mathbb{C}^*)^2$  is

$$f(t_1, t_2) = (1, t_1, t_2) \in \mathbb{CP}^2.$$

The image of  $f$  is dense, and its closure equals the whole projective plane. Hence

$$\mathbb{CP}_\Delta \cong \mathbb{CP}^2.$$

The normal fan  $\Sigma_\Delta$  consists of the cones generated by

$$(1, 0), \quad (0, 1), \quad (-1, -1),$$

which is exactly the standard fan of  $\mathbb{CP}^2$ . Thus this polytope reproduces the usual toric structure of the projective plane.

(2) *A scaled simplex: the Veronese embedding.*

Consider instead the scaled triangle

$$\Delta = \text{Conv}\{(0, 0), (2, 0), (0, 2)\}.$$

The shape of the polytope is unchanged, so its normal fan is the same as in part (1). Consequently the associated toric variety is again  $\mathbb{CP}^2$ .

However, the lattice points are now

$$\Delta \cap M = \{(0, 0), (1, 0), (2, 0), (0, 1), (1, 1), (0, 2)\},$$

six points in total.

Therefore the torus is embedded into  $\mathbb{CP}^5$  via

$$(t_1, t_2) \mapsto (1, t_1, t_1^2, t_2, t_1 t_2, t_2^2).$$

This map extends to all of  $\mathbb{CP}^2$  as

$$(x_1, x_2, x_3) \mapsto (x_1^2, x_1 x_2, x_2^2, x_1 x_3, x_2 x_3, x_3^2),$$

which is the classical Veronese embedding of  $\mathbb{CP}^2$  of degree 2.

Thus the same toric variety  $\mathbb{CP}^2$  can appear with different projective embeddings, depending on the choice of polytope.

(3) *Cutting a corner: blow-up of  $\mathbb{CP}^2$ .*

Now consider the quadrilateral

$$\Delta = \text{Conv}\{(0, 0), (2, 0), (1, 1), (0, 1)\}.$$

This polytope is obtained from the previous triangle by cutting off the corner at  $(0, 2)$ . Its lattice points are

$$\Delta \cap M = \{(0, 0), (1, 0), (2, 0), (0, 1), (1, 1)\}.$$

Using these five points, the torus embedding becomes

$$(t_1, t_2) \mapsto (1, t_1, t_1^2, t_2, t_1 t_2) \in \mathbb{CP}^4.$$

Attempting to extend this map to all of  $\mathbb{CP}^2$  gives

$$(x_1, x_2, x_3) \mapsto (x_1^2, x_1 x_2, x_2^2, x_1 x_3, x_2 x_3),$$

but this map is not defined at the point  $(0, 0, 1)$ .

To make the map well-defined, we must blow up  $\mathbb{CP}^2$  at this point. After performing the blow-up, the map becomes an embedding.

Hence the resulting toric variety is

$$\mathbb{CP}_\Delta \cong F_1,$$

the first Hirzebruch surface, which is the blow-up of  $\mathbb{CP}^2$  at one point.

Geometrically, cutting off a corner of the polytope corresponds to inserting a new ray in the fan, which is precisely the toric realization of the blow-up operation.

More generally, blowing up corresponds to cutting out an edge of  $\Delta$ .

We close this section with an easy way to picture the topology of  $\mathbb{CP}_\Delta$  directly from  $\Delta$ . We content ourselves with examples here.

For the first example, consider the map  $\mu : \mathbb{CP}^2 \rightarrow \mathbb{R}^2$  given by

$$(x_1, x_2, x_3) \mapsto \left( \frac{|x_2|^2}{|x_1|^2 + |x_2|^2 + |x_3|^2}, \frac{|x_3|^2}{|x_1|^2 + |x_2|^2 + |x_3|^2} \right).$$

The image of  $\mu$  is the set of all  $(a, b) \in \mathbb{R}^2$  satisfying  $a \geq 0$ ,  $b \geq 0$ ,  $a + b \leq 1$ , forming the triangle  $\Delta \subset \mathbb{R}^2$ . The fiber of a point in the interior of the triangle is a compact torus  $S^1 \times S^1$ , the fiber over an interior point of an edge is an  $S^1$ , and the fiber over a vertex is a point.

An even simpler example is  $\mathbb{CP}^1$  with bundle  $\mathcal{O}_{\mathbb{CP}^1}(1)$ , in which case  $\Delta$  is the interval  $[0, 1]$ . We have the map  $\mu : \mathbb{CP}^1 \rightarrow \mathbb{R}$  given by

$$(x_1, x_2) \mapsto \frac{|x_2|^2}{|x_1|^2 + |x_2|^2}.$$

The image of  $\mu$  is  $[0, 1]$ . The fiber of  $\mu$  over an interior point of  $[0, 1]$  is  $S^1$ , and the fiber of  $\mu$  over each of the endpoints is a point. This description leads immediately to the homeomorphism  $\mathbb{CP}_\Delta \cong S^2$ . The example of  $\mathbb{CP}^1$  is actually embedded in the example of  $\mathbb{CP}^2$  as the line  $x_3 = 0$ . Note that the bottom edge of the triangle in the left half of the figure can be identified with  $[0, 1]$ , compatibly with the lattice.

In general, there is a continuous map

$$\mu : \mathbb{CP}_\Delta \longrightarrow \Delta$$

such that the fiber of  $\mu$  over an interior point of a  $k$ -dimensional face of  $\Delta$  is homeomorphic to  $(S^1)^k$ .

### 3.6 Polytopes from Toric Varieties

Now we construct polytopes from projective toric varieties. Suppose we have a toric variety  $T \subset X$  embedded in a projective space  $\mathbb{CP}^k$ . This defines a hyperplane class  $\mathcal{O}_X(1)$  on  $X$ . We will need to assume that the action of  $T$  extends to an action on  $\mathbb{CP}^k$ , acting by coordinatewise multiplications.

To construct a polytope  $\Delta$ , we need to choose an isomorphism between  $\mathcal{O}_X(1)$  and  $\mathcal{O}(D)$ , where  $D$  is some fixed  $T$ -invariant divisor. Making a different choice for  $D$  will result in a translation of  $\Delta$ , so the choice of  $D$  is essentially irrelevant.

In the usual way, we identify sections of  $\mathcal{O}(D)$  with meromorphic functions  $f$  on  $X$  such that

$$(f) + D \geq 0,$$

where  $(f)$  is the divisor of  $f$ . Thus each coordinate function  $x_i$  on  $\mathbb{CP}^k$  is identified with a meromorphic function  $f_i$  on  $X$ . The condition that the  $T$ -action extends to  $\mathbb{CP}^k$  implies that the restriction of  $f_i$  to  $T$  is a character of  $T$ . We let  $m_i$  denote this restriction and identify it with an element of  $M$ . Then  $\Delta$  is the convex hull of the  $\{m_i\}$ .

**Remark 3.34.** In the construction of a polytope from a projective toric variety, one needs to choose a  $T$ -invariant divisor  $D$  and an isomorphism

$$\mathcal{O}_X(1) \cong \mathcal{O}(D).$$

This requirement often appears mysterious at first sight, and we briefly explain its necessity.

The goal is to extract from the projective embedding  $X \hookrightarrow \mathbb{CP}^k$  a collection of characters of the torus  $T \subset X$ , since the polytope  $\Delta \subset M_{\mathbb{R}}$  is defined as the convex hull of such characters. However, the homogeneous coordinates  $x_0, \dots, x_k$  on  $\mathbb{CP}^k$  are not global functions on  $X$ ; rather, they are global sections of the line bundle  $\mathcal{O}_X(1)$ . Hence they cannot be directly restricted to  $T$  as ordinary functions.

To convert these sections into meromorphic functions on  $X$ , we identify the hyperplane bundle with a divisor line bundle via an isomorphism  $\mathcal{O}_X(1) \cong \mathcal{O}(D)$ . Under this identification, each section  $x_i$  corresponds to a section of  $\mathcal{O}(D)$ , which in turn can be viewed as a meromorphic function  $f_i$  satisfying

$$(f_i) + D \geq 0.$$

Restricting  $f_i$  to the torus then yields a character  $m_i \in M = \text{Hom}(T, \mathbb{C}^*)$ . These characters are precisely the data used to define the polytope

$$\Delta = \text{Conv}\{m_0, \dots, m_k\}.$$

The existence of such a divisor  $D$  follows from general properties of toric varieties. For a projective toric variety  $X$ , every line bundle arises from a Cartier divisor, and in particular there exists a divisor  $D$  with

$$\mathcal{O}_X(1) \cong \mathcal{O}(D).$$

Moreover, one may always choose  $D$  to be  $T$ -invariant. Different choices of  $D$  correspond to linearly equivalent divisors, and hence change the resulting polytope only by a translation in  $M_{\mathbb{R}}$ . Since translations do not affect the normal fan, the associated toric variety remains unchanged. Therefore the choice of  $D$  is essentially irrelevant to the final geometry.

In summary, the divisor  $D$  serves as a bridge that allows homogeneous coordinates, which are sections of a line bundle, to be interpreted as meromorphic functions and hence as torus characters. This step is indispensable for translating the projective embedding data of  $X$  into the combinatorial data of a polytope.

**Remark 3.35.** The construction of a polytope depends on the choice of a line bundle, not merely on the toric variety  $X$ . More precisely, one should view the input as a pair  $(X, L)$ , where  $L$  is a  $T$ -linearized ample line bundle. If  $X$  is given together with a projective embedding  $X \hookrightarrow \mathbb{CP}^k$ , then this embedding determines a natural choice  $L = \mathcal{O}_X(1)$ . However, one may equally well replace  $\mathcal{O}_X(1)$  by any other ample line bundle  $L'$  (for instance,  $L' = -K_X$  when studying anticanonical hypersurfaces). The resulting polytope is then the polytope  $P_{D'}$  associated to  $L' \simeq \mathcal{O}_X(D')$ .

**Example 3.36.** We consider  $\mathbb{CP}^2$  with hyperplane class identified with  $\mathcal{O}(D_1)$ . The isomorphism between  $\mathcal{O}_X(1)$  and  $\mathcal{O}(D_1)$  is defined by division by  $x_1$ .

Thus, the coordinates  $\{x_1, x_2, x_3\}$  correspond to the meromorphic functions

$$\left(1, \frac{x_2}{x_1}, \frac{x_3}{x_1}\right)$$

respectively. The coordinates on the torus are given by

$$(t_1, t_2) = \left(\frac{x_2}{x_1}, \frac{x_3}{x_1}\right),$$

so the characters are  $(0, 0)$ ,  $(1, 0)$ , and  $(0, 1)$ , and we arrive at the polytope  $\Delta$  that leads us to  $\mathbb{CP}^2$ .

**Example 3.37.** For a toric variety  $X$  defined by a fan  $\Sigma$ , we have

$$\mathcal{O}(-K_X) \cong \mathcal{O}\left(\sum_{\rho \in \Sigma(1)} D_\rho\right).$$

In particular,

$$\mathcal{O}_{\mathbb{CP}^2}(3) \cong \mathcal{O}(-K_{\mathbb{CP}^2}) \cong \mathcal{O}(D_1 + D_2 + D_3),$$

where  $D_i$  is defined by the section  $x_i$  of  $\mathcal{O}_{\mathbb{CP}^2}(1)$ ,  $i = 1, 2, 3$ .

A basis of  $\Gamma(\mathcal{O}_{\mathbb{CP}^2}(3))$  is given by the ten homogeneous monomials of degree 3 in  $x_1, x_2, x_3$ .

Then with our choice

$$\mathcal{O}_{\mathbb{CP}^2}(3) \cong \mathcal{O}(D_1 + D_2 + D_3),$$

the degree 3 polynomial  $s$  is identified with the meromorphic function

$$\frac{s}{x_1 x_2 x_3}$$

on  $\mathbb{CP}^2$ . The  $T$ -action on the vector space

$$V = \left\{ \frac{s}{x_1 x_2 x_3} \mid s \in \Gamma(\mathcal{O}_{\mathbb{CP}^2}(3)) \right\}$$

has weights spanning the polytope  $\Delta \subset M_{\mathbb{R}}$ .

The cones over the proper faces of  $\Delta$  form a fan in  $M_{\mathbb{R}}$  in Figure 3, which we recognize as the fan of  $\mathbb{CP}^2/\mathbb{Z}_3$ . We will use this to illustrate mirror symmetry in the next section.

### 3.7 Mirror Symmetry

Batyrev has introduced a beautiful construction of mirror symmetry for Calabi–Yau hypersurfaces in toric varieties, based on the notion of duality for reflexive polytopes.

**Definition 3.38.** Let  $P$  be an  $n$ -dimensional polytope. A *facet* of  $P$  is a face of dimension  $n - 1$ , i.e. a face of codimension 1.

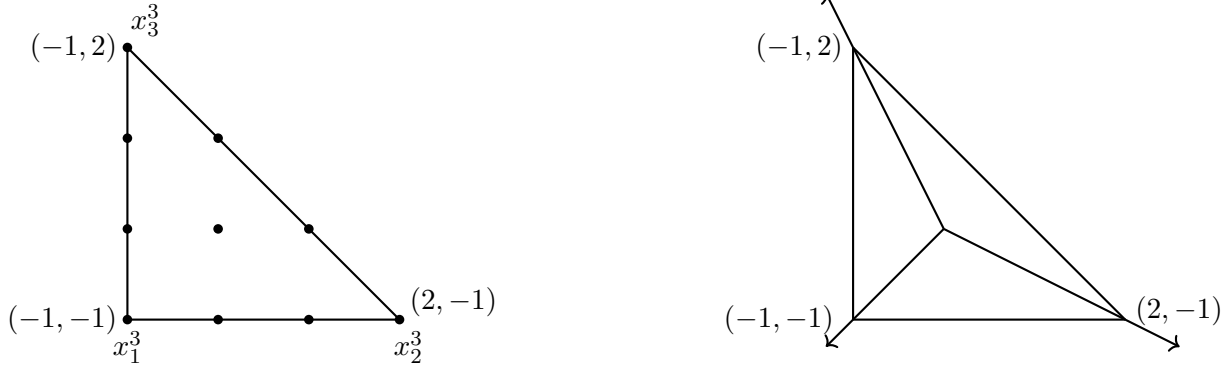


Figure 3: The polytope for  $\mathbb{CP}^2$  with bundle  $\mathcal{O}_{\mathbb{CP}^2}(3)$ .

**Definition 3.39.** A full dimensional lattice polytope  $P \subset M_{\mathbb{R}}$  is called *reflexive* if its facet presentation is of the form

$$P = \{ m \in M_{\mathbb{R}} \mid \langle m, u_F \rangle \geq -1 \text{ for all facets } F \},$$

where each  $u_F \in N$  is the primitive inward-pointing normal vector of the facet  $F$ .

**Proposition 3.40.** *Let  $P \subset M_{\mathbb{R}}$  be a full dimensional lattice polytope. Assume  $P$  is reflexive. Then  $0 \in M$  is an interior lattice point of  $P$ , and it is the unique interior lattice point of  $P$ .*

*Proof.* First, for every facet  $F$  we have  $\langle 0, u_F \rangle = 0 > -1$ , hence  $0 \in P^\circ$ .

Now let  $m \in M \cap P^\circ$  be any interior lattice point. Because  $m$  is interior, for every facet inequality we must have  $\langle m, u_F \rangle \geq 0$  for all facets  $F$ .

Assume for contradiction that  $m \neq 0$ . Then  $m$  defines a nonzero linear functional on  $N_{\mathbb{R}}$ , so there exists some vector  $v \in N_{\mathbb{R}}$  such that

$$\langle m, v \rangle < 0.$$

Since  $P$  is full dimensional, the primitive inward normals  $\{u_F\}$  of all facets span  $N_{\mathbb{R}}$ . Hence any vector in  $N_{\mathbb{R}}$  can be written as a nonnegative linear combination of these normals. In particular, the direction  $v$  must lie in the cone generated by some facet normal  $u_F$ , which implies

$$\langle m, u_F \rangle < 0$$

for some facet  $F$ .

This contradicts the fact that  $\langle m, u_F \rangle \geq 0$  for all facets  $F$ . Therefore  $m = 0$ .  $\square$

**Definition 3.41.** Let  $\Delta \subset M_{\mathbb{R}}$  be a full dimensional lattice polytope containing the origin in its interior. The *polar polytope* (or *dual polytope*) of  $\Delta$  is the polytope

$$\Delta^\circ = \{ n \in N_{\mathbb{R}} \mid \langle m, n \rangle \geq -1 \text{ for all } m \in \Delta \}.$$

Equivalently, if  $\Delta$  has facet presentation

$$\Delta = \{ m \in M_{\mathbb{R}} \mid \langle m, u_F \rangle \geq -1 \text{ for all facets } F \},$$

where  $u_F \in N$  are the primitive inward-pointing normals to the facets of  $\Delta$ , then the polar polytope  $\Delta^\circ$  is the convex hull of these normals:

$$\Delta^\circ = \text{Conv}\{ u_F \mid F \text{ is a facet of } \Delta \} \subset N_{\mathbb{R}}.$$

**Definition 3.42.** Let  $X$  be a normal algebraic variety. We say that  $X$  is *Gorenstein* if its canonical divisor  $K_X$  is Cartier, or equivalently, if the canonical sheaf  $\omega_X$  is a line bundle.

**Definition 3.43.** A projective variety  $X$  is called *Fano* if the anticanonical divisor  $-K_X$  is ample.

**Remark 3.44.** For a smooth variety  $X$ , the canonical bundle is defined as the line bundle of top degree holomorphic forms,

$$K_X = \Omega_X^n = \wedge^n \Omega_X^1.$$

However, when  $X$  has singularities, this definition no longer makes sense globally, since differential forms are only well-defined on the smooth locus.

For a normal variety, the canonical sheaf is defined in general by

$$\omega_X := j_* \Omega_{X_{\text{reg}}}^n,$$

where  $X_{\text{reg}}$  is the smooth part of  $X$  and  $j : X_{\text{reg}} \hookrightarrow X$  is the inclusion. This extends the usual notion of top degree forms from the smooth locus to the whole variety.

In general,  $\omega_X$  need not be a line bundle; it is only a reflexive sheaf. The variety  $X$  is called *Gorenstein* precisely when this canonical sheaf is invertible, i.e. when it becomes a genuine line bundle (equivalently, when the canonical divisor  $K_X$  is Cartier). Thus the Gorenstein condition ensures that the canonical bundle is well-defined even in the presence of singularities.

**Remark 3.45.** A Calabi–Yau variety is characterized by the stronger condition that the canonical line bundle is trivial, namely

$$K_X \cong \mathcal{O}_X.$$

Hence every Calabi–Yau variety is automatically Gorenstein, but the converse is false: the Gorenstein condition only requires  $K_X$  to be a line bundle, not to be trivial. In particular, a Gorenstein Fano variety is never Calabi–Yau, since for a Fano variety  $-K_X$  is ample and therefore  $K_X$  cannot be trivial.

**Theorem 3.46.** *Let  $\Delta$  be a full dimensional lattice polytope. Then the following statements hold:*

1.  $\Delta$  is reflexive if and only if the toric variety  $\mathbb{CP}_\Delta$  is Gorenstein and Fano.
2.  $\Delta$  is reflexive if and only if its polar polytope  $\Delta^\circ$  is reflexive.

Batyrev’s construction of mirror symmetry can be described as follows. Start with a reflexive polytope  $\Delta \subset M_{\mathbb{R}}$ . The normal fan  $\Sigma_\Delta$  of  $\Delta$  coincides with the fan obtained by taking the cones over the faces of the polar polytope  $\Delta^\circ$ .

The anticanonical divisor of the associated toric variety  $X_{\Sigma_\Delta} = \mathbb{CP}_\Delta$  is given by

$$-K_{X_{\Sigma_\Delta}} = \sum_{v_i \in \Sigma_\Delta(1)} D_i,$$

where the  $D_i$  are the torus invariant prime divisors corresponding to the rays of the fan. Sections of the anticanonical line bundle

$$\mathcal{O}_{X_{\Sigma_\Delta}} \left( \sum_{v_i \in \Sigma_\Delta(1)} D_i \right)$$

define anticanonical hypersurfaces in  $\mathbb{CP}_\Delta$ . These hypersurfaces are Calabi–Yau varieties, which we denote by

$$X \subset \mathbb{CP}_\Delta.$$

By Theorem 3.45, the polar polytope  $\Delta^\circ$  is also reflexive. Therefore the same construction can be applied to  $\Delta^\circ$ , producing a family of Calabi–Yau hypersurfaces

$$X^\circ \subset \mathbb{CP}_{\Delta^\circ}.$$

The assertion of Batyrev’s mirror symmetry is that the family  $X$  is mirror to the family  $X^\circ$ .

**Remark 3.47.** Although a projective toric variety  $X$  may be embedded into some projective space using the hyperplane bundle  $\mathcal{O}_X(1)$ , this embedding is not the data relevant for mirror symmetry. The objects of interest are not the toric varieties themselves, but the Calabi–Yau hypersurfaces

$$Y \in |-K_X|.$$

Mirror symmetry concerns the deformation theory of these hypersurfaces, whose complex moduli are governed by the space of sections  $H^0(X, -K_X)$ .

The polytope used in Batyrev’s construction is designed to encode precisely this linear system. For a line bundle  $L = \mathcal{O}_X(D)$  on a toric variety, the associated polytope is determined by the lattice points corresponding to sections of  $L$ . Hence, to describe the family of anticanonical hypersurfaces, the appropriate choice is the polytope

$$\Delta = P_{-K_X},$$

whose lattice points parametrize the monomials appearing in sections of  $-K_X$ .

If one were instead to construct a polytope using  $\mathcal{O}_X(1)$ , the resulting combinatorial data would encode the embedding of  $X$  itself, but would carry no information about the anticanonical linear system and therefore no information about the Calabi–Yau family. Thus the use of the polytope associated to  $-K_X$  is essential: it is this polytope that captures the geometry of the hypersurfaces whose mirrors are to be constructed.

**Remark 3.48.** The reason that a hypersurface defined by a section of  $-K_X$  is Calabi–Yau follows from the adjunction formula. Let  $X$  be a smooth projective variety and let

$$Y = Z(s) \subset X$$

be the zero locus of a section  $s \in H^0(X, L)$  of a line bundle  $L$ . Then the canonical bundle of  $Y$  is given by

$$K_Y \simeq (K_X + L)|_Y.$$

In the anticanonical case, we take  $L = -K_X$ . Hence for a hypersurface  $Y$  defined by a section of  $-K_X$  we obtain

$$K_Y \simeq (K_X + (-K_X))|_Y \simeq \mathcal{O}_Y.$$

Thus  $Y$  has trivial canonical bundle and satisfies the defining condition of a Calabi–Yau variety.

This explains why the anticanonical linear system plays a distinguished role: among all hypersurfaces in a toric variety, precisely those coming from sections of  $-K_X$  automatically produce Calabi–Yau manifolds.

**Theorem 3.49.** *If we use the standard embedding*

$$\mathbb{CP}_\Delta \hookrightarrow \mathbb{CP}^k, \quad k = |\Delta \cap M| - 1,$$

*then anticanonical hypersurfaces in  $\mathbb{CP}_\Delta$  are defined by linear equations in the homogeneous coordinates of  $\mathbb{CP}^k$ .*

In order to prove this, actually we need a useful lemma to write down the form of a section of the anticanonical bundle:

**Lemma 3.50.** *Let  $X_\Sigma$  be a toric variety, and let*

$$D = \sum a_i D_i$$

*be a torus invariant Cartier divisor on  $X_\Sigma$ , where the  $D_i$  are the torus invariant prime divisors corresponding to the rays of the fan  $\Sigma$ . Let  $\mathcal{O}_{X_\Sigma}(D)$  be the associated line bundle.*

*Define the polytope*

$$P_D = \{ m \in M_\mathbb{R} \mid \langle m, u_i \rangle \geq -a_i \text{ for all } i \}.$$

*Then the space of global sections of  $\mathcal{O}_{X_\Sigma}(D)$  is given by*

$$H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)) = \bigoplus_{m \in P_D \cap M} \mathbb{C} \cdot \chi^m.$$

*In particular, a basis of  $H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D))$  is naturally indexed by the lattice points of the polytope  $P_D$ .*

*Proof of Theorem 3.49.* The reason that anticanonical hypersurfaces in  $\mathbb{CP}_\Delta$  become linear equations after the standard embedding can be understood as follows. Let

$$\Delta \cap M = \{m_0, m_1, \dots, m_k\}$$

be the lattice points of the polytope. The standard toric embedding

$$\mathbb{CP}_\Delta \hookrightarrow \mathbb{CP}^k$$

is obtained by associating to each lattice point  $m_i$  the character  $\chi^{m_i}$ , so that the homogeneous coordinates of  $\mathbb{CP}^k$  correspond precisely to these monomials.

A section of the anticanonical bundle is of the form

$$s = \sum_{m \in \Delta \cap M} a_m \chi^m.$$

Under the above embedding, the monomials  $\chi^{m_i}$  become the homogeneous coordinates  $z_i$  of  $\mathbb{CP}^k$ . Hence the section  $s$  is represented in  $\mathbb{CP}^k$  by the linear expression

$$s = \sum_{i=0}^k a_i z_i.$$

Therefore its zero locus, which defines an anticanonical hypersurface in  $\mathbb{CP}_\Delta$ , is given by a linear equation in the homogeneous coordinates of  $\mathbb{CP}^k$ .  $\square$

**Remark 3.51.** In order to obtain the desired Calabi–Yau hypersurfaces, it is necessary to blow up the toric variety  $X_{\Sigma_{\Delta}}$  by performing a subdivision of the fan. In fact, a slightly stronger statement is required: there exists a subdivision of the fan for which the resulting blow-up is projective.

**Example 3.52.** We consider the classical example of quintic hypersurfaces in  $\mathbb{CP}^4$ . The mirror construction of Greene and Plesser describes the mirror family as the family of invariant quintic hypersurfaces in the quotient

$$\mathbb{CP}^4/\mathbb{Z}_5^3,$$

where  $\mathbb{Z}_5^3$  is the group of automorphisms of the form

$$(\alpha_1, \dots, \alpha_5), \quad \alpha_i^5 = 1, \quad \prod_{i=1}^5 \alpha_i = 1.$$

This description can be recovered using Batyrev’s construction. We start with the fan of  $\mathbb{CP}^4$ , whose rays are generated by the vectors

$$\begin{pmatrix} -1 & -1 & -1 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Since quintic hypersurfaces in  $\mathbb{CP}^4$  are anticanonical, the Batyrev mirror construction applies.

The dual polytope  $\Delta^\circ$  is the convex hull of the rows of this matrix. The set  $\Delta^\circ \cap N$  consists of six lattice points: the five rows above together with the origin, which we denote by  $v_0$ . These six lattice points determine an embedding

$$\mathbb{CP}_{\Delta^\circ} \subset \mathbb{CP}^5.$$

Let  $(y_0, \dots, y_5)$  be the homogeneous coordinates on  $\mathbb{CP}^5$ , where  $y_0$  corresponds to the origin  $v_0$ . From the relation

$$\sum_{i=1}^5 v_i = 5v_0,$$

we obtain the defining equation

$$y_1 y_2 y_3 y_4 y_5 = y_0^5$$

for the toric variety  $\mathbb{CP}_{\Delta^\circ}$ .

Under the change of coordinates

$$y_i = \hat{x}_i^5, \quad i = 1, \dots, 5, \quad y_0 = \hat{x}_1 \hat{x}_2 \hat{x}_3 \hat{x}_4 \hat{x}_5,$$

this equation is invariant under the automorphism group  $\mathbb{Z}_5^3$  described above. Hence this transformation induces an isomorphism

$$\mathbb{CP}_{\Delta^\circ} \cong \mathbb{CP}^4/\mathbb{Z}_5^3.$$

The anticanonical hypersurfaces in  $\mathbb{CP}_{\Delta^\circ}$  are given by linear expressions in the variables  $y_i$ . Under the above identification, these correspond precisely to  $\mathbb{Z}_5^3$ -invariant quintic hypersurfaces in  $\mathbb{CP}^4$ , as in the Greene–Plesser construction.

Alternatively, the identification  $\mathbb{CP}_{\Delta^\circ} \cong \mathbb{CP}^4/\mathbb{Z}_5^3$  can also be deduced by identifying the normal fan of  $\Delta^\circ$  with the fan obtained from the cones over the proper faces of the polytope  $\Delta$  corresponding to sections of  $\mathcal{O}(5)$  on  $\mathbb{CP}^4$ , following the general methods of toric geometry.

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